



Supplementary Materials for

Changing fitness effects of mutations through long-term bacterial evolution

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Materials and Methods

UMI-TnSeq approach

UMI-TnSeq Library Construction

We transformed *E. coli* MG1655 MFDpir with pSC189, a mobilizable plasmid that carries the mariner transposon but lacks the machinery necessary for replication outside the MFDpir host. We grew the donor strains in LB with 300 μ M DAP (diaminopimelic acid) and 50 mg/L kanamycin. We produced a fresh batch of donor strains each week, to ensure that the transposon had not been mobilized or disrupted essential genes in the donor background. We conjugated dense overnight cultures of the *E. coli* MFDpir-pSC189 donor and recipient (one of the LTEE strains) on a 0.2- μ m filter and incubated for 1 h. We resuspended the conjugation mixture and plated 1-2 mL on large LB agar plates supplemented with kanamycin (either 50 mg/L or 100 mg/L, depending on selection efficiency). After overnight growth at 30°C, we scraped cells from the plates, mixed the resulting mutant library with glycerol, and stored the library at -80°C . We plated dilutions of the transposon libraries and verified that they carried kanamycin resistance by patching ~ 50 colonies on LB and LB + kanamycin plates. We only retained those libraries where $>90\%$ colonies were kanamycin resistant.

UMI-TnSeq fitness assays

We performed bulk fitness assays in DM25 (Davis-Mingioli minimal medium with 25 mg/L glucose) held in glass test tubes and incubated at 37°C with 220 rpm orbital shaking. We used test tubes, instead of the small Erlenmeyer flasks used in the LTEE, in order to run many fitness assays simultaneously. Previous work found no systematic difference in fitness estimates using glass tubes versus flasks. Each fitness assay comprised five 10-mL cultures propagated in parallel. This replication increased the bottleneck population size by a factor of five, without introducing any density-dependent effects. For each assay, we inoculated 50 mL of DM25 with $\sim 5 \times 10^5$ cells/mL from the freezer stock of the transposon library (comparable to the number of cells transferred during the LTEE), and we then split the volume among five glass tubes. After incubating for 24 h at 37°C with shaking, we pooled the five tubes. For four days (~ 26.5 generations in total) we diluted 500 mL of each bulk competition into 50 mL fresh DM25 (1:100 daily dilution, as in the LTEE) and immediately split the volume among five cultures, as before. We spun down the remaining cells, and we stored the pellet frozen for later DNA extraction and analyses. For each transposon library, we performed two replicate fitness assays, starting from the same aliquot of the transposon library from the freezer stock (which served as the time-zero point).

Recombineering and Pairwise Fitness Assays

As a control, single gene deletions were constructed by recombineering, as previously described (79). We used the pSIM5-tet plasmid carrying a heat-inducible lambda red recombinase and a tetracycline-resistance gene (80). The kanamycin-resistance cassette used for recombineering was amplified from the pKD4 plasmid (79). We grew overnight cultures of the strains for recombineering (containing the pSIM5-tet plasmid) in LB at 30°C, and we diluted them 100-fold in 50 mL of low-salt LB. When the cultures were in mid-exponential phase (OD ~ 0.4), we transferred them to a water bath at 42°C to induce the recombinase for 15 min. We then chilled the cultures on ice, and we prepared competent cells from the induced cultures by washing and pelleting cells multiple times in 10% glycerol. We then transformed the competent cells with 200-

500 ng of the recombineering insert, which had 40 bp identical to the flanking regions of the gene we sought to delete as well as the kanamycin- resistance cassette (primer sequences shown in Table S5). The recombineering protocol worked successfully for the LTEE ancestor (REL606), but we were unable to make deletion mutations in the evolved strains except the 50,000-generation clone from population Ara—1 (REL11330), for which we saw a reduced efficiency. We found that the limiting factor for most evolved strains was transformation of the recombineering plasmid into the cell using electroporation.

For pairwise competitions, we grew overnight cultures of the unmutated parent and the deletion mutant in LB. We mixed them 1:1 volumetrically and inoculated the mix in 10 mL of DM25 with $\sim 5 \times 10^5$ cells/mL. We spread dilutions from the competition mixture on LB and LB + Kanamycin plates at the start of the competition and after one day in DM25. The counts on the LB + Kanamycin plates correspond to the number of mutant cells, while the counts on the LB plates correspond to the total number of cells including both the parents and mutants. We estimated the fitness effect of the deletion mutation as the rate of change in the ratio of the mutants to the parents, as follows:

$$s = \ln \frac{N_{LB+Kan,t1} / (N_{LB,t1} - N_{LB+Kan,t1})}{N_{LB+Kan,t0} / (N_{LB,t0} - N_{LB+Kan,t0})} / \log_2(100)$$

where $\log_2(100)$ is the number of generations (doublings) per day during the competition given the 100-fold dilutions.

UMI-TnSeq Library Preparation and Sequencing

We used the Invitrogen PureLink gDNA extraction kit to extract DNA from $\sim 2 \times 10^9$ cells for each transposon library. We measured DNA concentrations using the Invitrogen Quant-it Kit and normalized them to 20 ng/ μ L. We ran 10 μ L tagmentation reactions with 5 μ L Illumina TDE1 buffer, 2.5 μ L TDE1 enzyme, and 2.5 μ L normalized DNA at 55°C for 10 min. We used the entire volume of the reaction as the template for PCR1, with 3 μ L each of 10 μ M stocks forward primer TnSeq_1F and reverse primer TnSeq_1R_xx (see Table S6 for sequences), and 1.5 μ L of Q5 2X master mix (New England Biolabs) with the following reaction cycle: first 5 min at 95 °C; 3 cycles of 30 s at 98 °C, 30 s at 68 °C, 1 min at 72 °C; and finally 3 min at 72 °C, followed by storage at 4 °C. We cleaned the product of PCR1 with 1.2X serapure beads, which we eluted in 15 μ L dH₂O. We used the eluant as template for PCR2, where we selectively amplified fragments containing the transposon sequence, with 5 μ L each of 10 μ M stocks forward primer TnSeq_2F_yy and reverse primer TnSeq_2R (Table S6) and 25 μ L of Q5 2X master mix with the following reaction cycle: first 5 min at 95 °C; 19 cycles of 30 s at 98 °C, 30 s at 68 °C, 1 min at 72 °C; and finally 3 min at 72 °C, followed by storage at 4 °C. Primer combinations used in the study are shown in Table S7. We cleaned the product of PCR2 with 1.2X serapure beads, which we eluted in 25 μ L dH₂O. We quantified the concentration of DNA of the amplified transposon libraries using the Invitrogen Quant-it Kit, and we diluted and pooled all samples to 4 nM. We verified the concentration of the pooled library using the Kapa Biosystems Illumina qPCR kit. We also prepared whole-genome sequencing libraries for all the strains using the tagmentation based-approach from Baym *et al* (81). We sequenced the transposon and whole-genome libraries on two lanes of an Illumina NovaSeq S4 (paired end, 2 x 150 bp) at the Bauer Core Facility, Harvard University.

UMI-TnSeq data analyses: Fitness and Essentiality estimates

We used the Invitrogen PureLink gDNA extraction kit to extract DNA from $\sim 2 \times 10^9$ cells for each transposon library. We measured DNA concentrations using the Invitrogen Quant-it Kit and normalized them to 20 ng/ μ L. We ran 10 μ L tagmentation reactions with 5 μ L Illumina TDE1 buffer, 2.5 μ L TDE1 enzyme, and 2.5 μ L normalized DNA at 55°C for 10 min. We used the entire volume of the reaction as the template for PCR1, a low-amplification cycle PCR where we added unique molecular identifiers using custom primers (see Part 4 of Detailed Experimental Protocols for primer sequences). We cleaned the product of PCR1 with 1.2X serapure beads, which we eluted in 15 μ L dH₂O. We used the eluant as template for PCR2, where we selectively amplified fragments containing the transposon sequence. We cleaned the product of PCR2 with 1.2X serapure beads, which we eluted in 25 μ L dH₂O. We quantified the concentration of DNA of the amplified transposon libraries using the Invitrogen Quant-it Kit, and we diluted and pooled all samples to 4 nM. We verified the concentration of the pooled library using the Kapa Biosystems Illumina qPCR kit. We also prepared whole-genome sequencing libraries for all the strains using the tagmentation based-approach from Baym *et al* (81). Additional details on the UMI-TnSeq library preparation are included in Part 3 of the Detailed Experimental Protocols. We sequenced the transposon and whole-genome libraries on two lanes of an Illumina NovaSeq S4 (paired end, 2 x 150 bp) at the Bauer Core Facility, Harvard University.

Data Analysis: Fitness

We obtained demultiplexed reads from the sequencing core. We filtered reads using a custom Python script, retaining only those reads that contained a sequence matching the end of the mariner transposon, and we stored the 10-bp unique molecular identifier (UMI) sequence separately. We used bowtie2 to map the filtered reads to the reference genome of the LTEE ancestor, *E. coli* strain REL606. We extracted coordinates of all the uniquely mapped reads; these coordinates correspond to the TA sites in the reference genome. For every TA insertion site, we identified corresponding UMIs and mapping coordinates, and we used this combination to count the number of distinct biological replicates for each TA site. We obtained a median of 23 million reads per timepoint in the fitness assays. Approximately 85% of the reads mapped to the reference genome; $\sim 20\%$ of those reads were discarded as PCR duplicates. We consolidated all the data from the bulk fitness assay into a master file for each transposon library; that dataset includes the counts for every insertion site at each timepoint in the fitness assay.

For the downstream analyses, we normalized each insertion by the total sample size. For each insertion mutation with at least 5X coverage at the start of the assay, we estimated its fitness as the slope of the linear fit to its $\ln(\text{frequency})$ over time (in number of generations, $\log_2(100) \approx 6.64$ per day). As explained elsewhere, this metric differs from the ratio of Malthusian growth rates (as used in many other LTEE analyses) by a factor of $\ln 2$ (82). For those mutants that disappeared, whether due to chance or the mutant being deleterious, we added a pseudo-count of 1. For each gene, we then averaged over all the insertion sites using an inverse-variance weighting approach, excluding the first 10% and last 25% of the gene's length.

We estimated the error in the fitness measurement of a gene as the weighted standard error of the mean of the fitness estimates for all K interior TA sites within the gene:

$$\mu_{\text{gene}} = \left(\frac{1}{K}\right) \sum_{i \in \text{TA sites}} w_i S_i.$$

The weights were defined as follows:

$$w_i = \frac{\log(2)^2}{\left(\frac{1}{1+n_0} + \frac{1}{1+n_1}\right)},$$

where n_0 and n_1 are the number of reads at timepoints 0 and 1, respectively, and K is the number of TA sites. Maximum weights were set with n_0 and n_1 both equal to 100. We define the error in the fitness estimate of a gene as the inverse-variance weighted standard error of the mean:

$$\Delta\mu_{gene} = \sqrt{\frac{1}{(N)(N-1)} (\sum_{i \in TA \text{ sites}} w_i (S_i - \mu_{gene})^2)}.$$

The fitness effect of each gene was adjusted by a correction factor equal to the average fitness effect of insertions that disrupt pseudogenes. This adjustment ensures that our fitness estimates are not influenced by changes in the mean population fitness during the bulk fitness assay. The mean of the DFE is:

$$\mu_{DFE} = \sum_{genes} \mu_{gene} / N - \sum_{pseudogenes} \mu_{pseudogene} / M,$$

where N and M are the numbers of genes and pseudogenes, respectively. We define the uncertainty in the estimate of the mean of the DFE as:

$$\Delta\mu_{DFE} = \sqrt{\sum_{genes} (\Delta\mu_{gene} \frac{\partial \mu_{DFE}}{\partial \mu_{gene}})^2 + \sum_{pseudogenes} (\Delta\mu_{pseudogene} \frac{\partial \mu_{DFE}}{\partial \mu_{pseudogene}})^2}, \text{ where}$$

$$\frac{\partial \mu_{DFE}}{\partial \mu_{gene}} = 1/N, \frac{\partial \mu_{DFE}}{\partial \mu_{pseudogene}} = -1/M.$$

We restricted our analysis to those protein-coding genes with at least 5 interior TA sites; we further required that a gene had at least two trajectories meeting thresholds in both technical replicates, and that at least 20% of the gene's TA sites were used in the fitness estimation. We chose these thresholds to ensure that we had sufficient data for every gene that was included in our analyses of changes in essentiality, and to exclude potential outlier sites in essential genes from contributing to the fitness estimates. We realize that excluding 10% and 25%, respectively, of the 5' and 3' ends of a gene is an imperfect solution, but it greatly reduced the impact of outliers. We also excluded all genes with annotations linked to transposons or mobile genetic elements, because they might move between sites in the genome. When comparing fitness estimates between two strains, we often found that fitness effects were calculated in one strain but not the other owing to these arbitrary thresholds. To get around this problem, we calculated fitness effects in such cases using relaxed thresholds (i.e., at least one trajectory meeting thresholds in both replicates, instead of two as otherwise required).

Data Analysis: Essentiality

Differential essentiality in DM25

To classify a gene as differentially essential between ancestral and evolved strains in DM25, we required mutations in that gene to have a fitness effect > -0.15 in one strain and < -0.3 in the other. The inclusion of the -0.15 threshold was to ensure that the difference in fitness effects between strains is sufficiently large that it cannot be attributed to measurement noise. The choice of the -0.3 threshold is based on numerical simulations of fitness effects for genes that are essential for growth in DM25, which we used to identify the most deleterious fitness effect such that the overlap

in the distributions for an essential gene and a non-essential gene subject to the merely deleterious mutations is < 0.05 (Fig. S8).

The simulations were performed as follows:

- Fit a distribution to the number of reads per TA site from experimental data using the ANC strain at the initial timepoint t_0 (restrict to sites with at least 10 reads). We fit both normal and gamma distributions to the log-transformed number of reads per TA site. Both distributions work well, but we report results using the gamma distribution.
- Draw an initial abundance from the fit to experimental data, which is the number of reads at t_0 and called N_0 .
- Estimate the expected number of reads at t_1 as follows:
 - $E[N_1] = N_0/100 * \exp((1 + s)6.64) = N_0 * \exp(6.64s)$ if the gene is not essential. The value 6.64 equals the number of generations (doublings) of population growth per day, which offsets the 100-fold daily dilution.
 - $E[N_1] = N_0/100$, if the gene is essential, such that the mutant does not grow at all. The division by 100 reflects the daily 100-fold dilution without growth.
 - Draw from a Poisson distribution with mean $E[N_1]$, which we call N_1 .
- Estimate the simulated fitness effect as:
 - $s = \ln((N_1 + 1)/N_0 + 1))/6.64$.
- Repeat the simulation 5,000 times.

Differential essentiality in LB

Even with relaxed thresholds for fitness estimation, there were many cases of genes that were nonessential in DM25 in one strain but did not have any estimated fitness in the other strain. To determine whether these cases arose because insertion sites were missing by chance, or whether the gene was in fact differentially essential in LB, we used the following approach:

- We first define essentiality in LB based on the fraction of TA sites, f , in a gene that had at least 5 reads mapping to a site at time 0 (after correcting for sequencing depth). We classified those genes with $f < 0.1$ as essential in LB (Fig. S8A).
- For the LTEE ancestor REL606, this definition captured 88% of the essential genes in *E. coli* K-12 (48), with a false positive rate of only 2%. The area of the receiver-operator characteristic curve is 0.96 (Fig. S8B), which validates our approach to defining a gene as essential in LB.
- Next, identify all genes that are nonessential ($s > -0.15$) in DM25 in strain 1, and with fitness not calculated in strain 2, with fraction $f < 0.1$.
- For every such gene, calculate the number of TA sites at time 0 with coverage $> 5X$ in both strain 1 (n_1) and strain 2 (n_2).
- Calculate the probability of observing n_2 or fewer sites, assuming that the expected number of sites is n_1 .
- Adjust those p -values using the Benjamini-Hochberg False Discovery Rate (FDR).
- Define those genes for which this FDR-adjusted probability is < 0.05 as differentially essential in LB for the two strains in question.

This approach ensures that we are conservative in calling genes differentially essential, particularly in those genes with fewer than 10 insertion sites. As an additional control, using the same parameters as described above, we did not find any genes that are differentially essential between

the two marked variants of the LTEE ancestors (strains REL606 and REL607, which we call Anc and Anc*, respectively) that have repeatedly been shown to have equal fitness.

Estimating the number of essential genes in the ancestor in DM25

We counted the number of genes where disruptions led to a fitness estimate of $s < -0.3$. To account for the fact that some genes may be missing from the initial transposon library by chance (i.e., due to very few TA sites within the gene's interior), we excluded those genes with an FDR-corrected probability > 0.05 (as described for defining differential essentiality in LB).

UMI-TnSeq data analyses: Structural Variation

The average depth of coverage of the whole-genome sequencing data was $> 60X$. We used breseq (83) to identify regions of the genome that had been deleted in the evolved strains during the LTEE. We identified deletions > 1 kb using the missing-coverage evidence from the breseq output files. To identify duplicated regions, we used the samtools depth command (84) to get depth of coverage at every site in the genome, using the ancestral REL606 genome as the reference. To account for variability in coverage across the genome, we calculated the normalized coverage for each evolved strain relative to REL606. We obtained the coordinates for duplicated genes by visualizing the normalized coverage and identifying those regions where the coverage was consistently 1.5X or higher than the background level. We set the cutoff at $> 1.5X$ coverage over background because large duplications also have a tendency to be lost when clones are being grown, as is done to obtain their DNA for sequencing.

To identify gene homologs, we used MMseqs2 cluster (with parameters: --min-seq-id 0.4) with a 40% identity threshold for homologs (85). For each evolved clone, we asked whether all but one member of the paralogous group had been lost during evolution and, if so, we evaluated whether the other member had become essential.

INSeq approach

INSeq Library Construction

We employed a kanamycin-resistant version of the original INSeq transposon carried by the pSAM plasmid (76). The pSAM plasmid also carries the Himar1C9 transposase, an ampicillin-resistance marker, and RP4-oriT/oriR6K, a mobilizable and *pir*-dependent origin of replication. We maintained the plasmid in *E. coli* MG1655 MFD*pir* (75); from there, it was transferred to the *E. coli* recipient strains by standard agar-plate conjugation methods. Mating time was limited to 3-4 h, after which we selected transconjugants by spreading the cells onto LB agar plates supplemented with streptomycin (100 mg/L) and kanamycin (100 mg/L). Libraries were assembled from at least 100,000 colonies, which were obtained from at least 10 independent conjugation mixtures. The colonies were scraped off the plates, pooled in saline solution, and glycerol was added (10% final) for storage at -80°C .

INSeq fitness assays

We propagated aliquots of the mutant libraries in the same conditions as those used in the LTEE: 1/100 daily dilutions in 50-mL flasks filled with 10 mL of DM25 and incubated at 37°C in an orbital shaker (47). To ensure that the initial cell density was similar to that used in the LTEE ($\sim 5 \times 10^7$ cells/mL), we inoculated several flasks for each experiment with a different aliquot from the

thawed libraries, using only those with appropriate densities based on viable cell counts (and discarding the others). The competition experiments ran for 5 or 8 days (Ara+2 and Ara-1 samples, respectively); we stored the remaining culture after each serial passage at -80°C for later analyses. We monitored the populations based on viable cell counts to verify that they grew for the expected ~ 6.64 generations each day.

INSeq Library Preparation and Sequencing

The thawed stocks from the bulk competition experiments were pelleted and subjected to genomic DNA extraction using the DNeasy Blood & Tissue kit (Qiagen). Owing to the relatively low cell density supported by DM25, the purified DNA was subjected to high-fidelity, unbiased whole-genome amplification using the *Phi29*-polymerase-based REPLI-g midi kit (Qiagen). The resulting DNA was digested by *MmeI* (NEB), concentrated using a vacuum concentrator, and run on a 1% agarose gel. The expected size of the transposon plus the adjacent genomic DNA is ~ 1.5 Kb, so we excised from the gel the corresponding region, using as a reference the 1.2-1.5 Kb bands from the Quick-Load 100 bp DNA ladder (NEB). To enrich the excised samples for transposons, the INSeq method takes advantage of the fact that *MmeI* digestion results in dinucleotide, sticky-end fragments; custom-made PCR adaptors could then be added in a simple ligation reaction (Fig. S19, Table S8). We performed the ligations using T4 DNA ligase (NEB). Ligation products contain the 14 bp adjacent to the insertion site flanked by the transposon and the custom-made adaptor. Using these known sequences for annealing, we used a 28-cycle PCR to amplify a small region containing the 14 bp of interest (Fig. S19, Table S8). All PCR reactions were carried out using the HiFi DNA Polymerase (KAPA Biosystems). The resulting molecules (116 bp and 148 bp in size for the Ara+2 and Ara-1 samples, respectively) were isolated by electrophoresis and band excision on a 2% agarose gel. To add the required indices for multiplexed, paired-end sequencing, we subjected the samples to a secondary, 18-cycle PCR using the Nextera XT DNA Library Prep Kit (Illumina). The final Nextera libraries were purified, quantified, and sent for sequencing on Illumina platforms. The Ara+2 libraries were sequenced using a HiSeq platform (Integrage, Evry, France), while the Ara-1 libraries were sequenced using a MiSeq platform (Bichat Hospital, Paris, France). After filtering (see below), the libraries had on average ~ 3.55 and ~ 0.39 million reads, mapping to an average of ~ 0.33 and ~ 0.27 million different insertion sites for Ara+2 and Ara-1, respectively. To control for potential technical biases, libraries for the ancestor were processed using both transposon capture and sequencing methods (Fig. S20).

INSeq data analyses: Fitness estimates and outlier removal

We developed a series of custom scripts to map transposon insertion sites, discard low-quality cases, and quantify the abundance of each insertion allele. We first used a Python script to extract the captured 14-bp genomic fragments from the Illumina sequence reads. This step was accomplished using the module “regex” to identify any of the constant sequences expected to flank the captured fragments, up to a maximum edit distance of 1 (*i.e.*, one position indel or mismatch) (Fig. S19). Second, we used the BWA algorithm (86) to map these genomic fragments to the *E. coli* B str. REL606 reference genome (NCBI reference: NC_012967), again with a maximum edit distance of 1. We only retained fragments for which the best hit mapped to a single genomic location, thus excluding repeated regions from the analysis. Pairs from paired-end reads were merged except when they gave discordant results (in these cases, only the one with the best alignment was retained). These filtered read mappings were then compiled in a list with all the unique insertion sites, the number of reads per site and the identity of the corresponding locus. As

loci, we considered both open reading frames and intergenic regions, but we excluded repetitive and mobile genetic elements. Finally, to characterize sub-genic effects (Fig. S2), we divided each locus into equally sized segments (5 for the Ara+2 dataset, and 3 for the lower-coverage Ara-1 dataset; in both cases, loci were not divided when <100 bp in size), and we calculated the abundance of all insertions that mapped onto the same segment.

In preliminary tests we observed that excluding insertion sites with very low counts reduced the number of artifacts in downstream analyses. This effect presumably occurs because very low abundance insertions are enriched for PCR and Illumina sequencing artifacts. Therefore, for each segment, we pooled together only insertions with >10 or >3 total counts across all timepoints (Ara+2 and Ara-1, respectively). All the following analyses were conducted using the data for these pooled segments. As an added benefit, comparing the fitness effects among segments of the same locus allows identification of potential artifacts and provides an internal control to quantify the reproducibility of the fitness estimates (Fig. S14).

To estimate the fitness of the pooled insertion mutants for each gene or sub-genic region, we calculated the slope of their frequency trajectories relative to that of a set of presumed neutral loci. The natural logarithm of this slope approximates the selection coefficient from classical population genetics, and it is related to the ratio of realized growth rates (the standard fitness metric used in the LTEE literature) by a scaling factor of $\ln 2$ (Fig. S22A) (82). The set of presumed neutral loci includes genes annotated as cryptic and the L-arabinose transport and catabolism operons (L-arabinose auxotrophy has been widely used as a neutral marker in the LTEE). In addition, we only considered as neutral their internal segments, in order to avoid spurious deviations from neutrality associated with polar or change-of-function effects (e.g., Fig. S2). We estimated slopes by fitting a linear regression model to the data using the `lm()` function in R. We weighted the regression using the counts at each time point (using the “weight” argument), in order to reduce the noise associated with low counts. We found this weighting improved estimates in the deleterious range (Fig. S22B). Mutants with fewer than 3 non-zero timepoints were discarded.

In tests with the neutral set, we observed that most poor-quality fits could be filtered out simply by excluding mutants with very low initial abundance (cutoffs for exclusion of <10 or <3 counts at the first timepoint for Ara+2 and Ara-1, respectively). However, when the much larger complete set of loci was analyzed, we noticed several instances in which marked outliers were characterized by increasingly steep slopes at later timepoints. These cases probably reflect hitchhiking of individual insertion alleles with *de novo* beneficial mutations that arose during the competition experiments. Consistent with this interpretation, these outliers were proportionately more common in the higher-coverage Ara+2 dataset, presumably because of the increased resolution and hence ability to detect these rare events. To remove these outliers, we filtered them out using the standard error of the regression (cutoffs of 0.01, all cases).

We also noticed a few highly beneficial mutations in the ancestor from the Ara+2 dataset that we suspect constituted another class of more subtle artifacts. Our suspicion arose from two facts: first, they were completely absent from the ancestor data for Ara-1; second, their fitness markedly deviated from the fitness measured for the other insertions in the same locus. One explanation for these artifacts is that they arose from some type of PCR bias. However, MmeI digestion ensures that genomic fragments are of uniform length (1.5 Kb), thus preventing any significant size-related

extension rate differences across fragments (87). Other types of PCR bias cannot be excluded, but they probably play a minor role, as shown in a recent study comparing PCR-free versus PCR-based transposon enrichment methods (88).

Alternatively, we think the likely explanation is that these artifacts represent even rarer hitchhiking events, ones in which an insertion occurred in a genetic background harboring either a preexisting beneficial mutation or a secondary beneficial insertion. In such cases, the insertion would behave as a beneficial mutation over the entire time course, and thus it cannot be identified and discarded based on a high standard error of the regression. These rare events could even make deleterious mutations appear neutral or beneficial. To address this issue, we systematically compared all segments for each locus, and we removed those internal segments that deviated by > 0.01 in fitness from the locus average (except in cases where the deviation represents $< 25\%$ of the locus average, to avoid being too strict with loci displaying large deleterious effects; see Fig. S14B). We retained N- or C-terminal segments, on the grounds that such deviations may represent polar or change-of-function effects; however, to retain them when beneficial, we further required that other beneficial mutations were found elsewhere in the same operon.

Overall, after imposing all of these filters for the Ara+2 dataset, loci retained an average of 3.5 ± 1.3 internal segments (mean $\pm$ SD; Anc: 3.2 ± 1.3 , 2K: 3.7 ± 1.3 , 15K: 3.6 ± 1.3), with each internal segment having an average of 7.2 ± 5.8 independent insertion alleles (Anc: 7.8 ± 6.3 , 2K: 7.6 ± 6.1 , 15K: 6.1 ± 4.8). For the Ara-1 dataset, loci retained an average of 2.3 ± 0.8 internal segments (mean $\pm$ SD; Anc: 2.3 ± 0.8 , 2K: 2.3 ± 0.8 , 15K: 2.2 ± 0.8), with each internal segment having on average 7.4 ± 5.8 independent insertion alleles (Anc: 8.3 ± 6.8 , 2K: 7.5 ± 5.9 , 15K: 6.3 ± 4.8).

INSeq data analyses: Statistical methods

We performed all statistical tests using either built-in functions or publicly available packages from the R programming language (version 3.6.3). To fit linear models, we used the built-in `lm()` function. We used the package “Matching” to implement a bootstrap version of the Kolmogorov-Smirnov test, which corrects for the fact that the empirical distributions of fitness effects are discrete in nature (89). We used the package “Fitdistrplus” to find which of the distributions commonly used in the literature provided the best fit to the beneficial tails (Table S2) (90). For all tests conducted on both the deleterious and beneficial tails, we included only those values outside the range of neutrality, as established from the DFE for the set of presumed neutral loci (Fig. S23). To assess whether measurement error might drive the observed results, we resampled the data using the logarithm of the regression standard errors as weights. Weighted resampling with replacement was conducted 200 times for each genetic background and tail of the distribution, and the resulting datasets were used to repeat the tests associated with Figures 2B and 2C. These analyses confirmed that our main results hold even when accounting for measurement error, both for the deleterious and beneficial tails (Fig. S24). We also used the resampled datasets to estimate 90% confidence intervals around the cumulative DFEs (Fig. 5B).

To test whether beneficial mutations in the ancestor tended to cluster together in operons more often than expected by chance, we sampled without replacement an equal number of loci from the reference genome and computed the frequency of having another member from the same transcription unit in the sample. The reported p-value is the proportion of cases in 100 simulations

that show an equal or higher frequency of clustered loci than the observed data. Operons in which mutations typically have small fitness effects should be underrepresented because most mutations will fall below the threshold of neutrality. To avoid this potential bias, we focused these analyses on the loci that harbored the 50% most beneficial mutations observed. The list of transcriptional units for the ancestor was retrieved from the BioCyc database (biocyc.org/group?id=:ALL-TRANSCRIPTION-UNITS&orgid=ECOL413997, accessed on 2/23/2022).

INSeq data analyses: DFE-based predictions versus allele prevalence in the LTEE

We obtained the metagenomic time-series data for the LTEE from <https://github.com/benjaminhgood/>; genomic positions were used to standardize annotated loci labels as needed. For every non-synonymous allele observed in the metagenomic data, we recorded the number of lineages in which a similar allele was detected, the maximum frequency it reached, and the timepoint at which this maximum was attained. By similar allele, we mean any other non-synonymous located in the same gene or sub-genic region. Six of the LTEE lineages evolved hypermutability at various generations, which introduced many “passenger” mutations (i.e., neutral or mildly deleterious mutations that reach high frequency due to their linkage with a beneficial “driver” mutation) (91). Such passengers are typically unique because they are drawn from a very large pool of potentially neutral or mildly deleterious mutations. Therefore, we restricted our analyses to similar alleles (as defined above) that arose in at least 2 independent lineages. In addition, to ensure that these alleles represented beneficial mutations, we only considered those alleles that reached a frequency of at least 50% at some point in the corresponding metagenomic trajectory. When attempting to correlate the abundance of these presumptive driver alleles with our fitness data, we realized that working at the sub-genic and even the locus level relied on sparse data and thus had low statistical power. To overcome this limitation, we performed the analyses at the operon level; fitness was computed as the maximum value recorded for any of its constituents (including sub-genic regions), and target size was calculated as the sum of relevant constituent lengths (i.e., excluding sub-genic regions and loci without beneficial effects).

We followed the same approach when we looked at how well the DFEs predicted subsequent steps of adaptation. To compute the fraction of drivers that were captured by the DFE, we retrieved the alleles reaching $\geq 50\%$ frequency at a given timepoint and counted how many of them belonged to operons present in the beneficial tail of the corresponding DFE. In the analyses with the ancestor, we required the allele to reach the 50% threshold in at least 2 LTEE lineages. In the same analyses, except performed with the 2K and 15K genetic backgrounds, we only considered alleles that surpassed the threshold in their own lineage (Ara+2 or Ara-1), discarding those alleles already present above this threshold at 2K or 15K generations, respectively. Note that despite the focus on a single lineage, the mutations still come from the filtered “driver” set; hence, lineage-specific singletons were not considered. To compute a null expectation for these analyses, we ran them on random samples from the neutral and deleterious fractions of each DFEs. The samples were drawn without replacement and with a size equal to the beneficial mutation set to which they were compared. The shaded areas in Figures 6, S16, S18 and S20 show the average of 50 simulations $\pm$ three times the standard error of the mean.

Supplementary Text

1. Exclusion of Ara-2 from further analysis

We excluded the Ara-2 sample from all comparisons after we found that fitness measurements in this clone were unreliable and systematically biased. In this genetic background, a few insertion mutations massively increased in frequency during the fitness assay, depressing the frequencies of other mutations. In principle, we might compensate for this effect by excluding the outliers, and then correcting the other mutations using pseudogenes as the neutral expectation. However, the insertion mutations in pseudogenes were also outcompeted in the fitness assay; their counts fell to nearly zero on day 4, and the estimated fitness effects for these insertions range from -0.2 to -0.15 . As a consequence, there are two unavoidable problems with using the pseudogenes to establish a correction factor for the remainder of the DFE:

- Measurements of deleterious fitness effects are inherently noisier than nearly neutral effects, leading to generally larger errors and potentially systematic biases for fitness estimates of all other mutations if we used the mean pseudogene value as a correction factor.
- By subtracting a large, constant deleterious fitness effect (nearly -0.2) from all other measurements, the distribution of fitness effects in Ara-2 would abruptly end at $s \approx -0.3$. This approach would effectively truncate the deleterious tail of the DFE, which in other backgrounds spans effects as deleterious as -0.6 to -0.3 . Consequently, a calculated fitness effect of -0.25 , for example, could not be meaningfully compared between Ara-2 and other populations.

2. Gain or loss of functional redundancy explains some, but not most, changes in essentiality

Gene essentiality may also evolve as by-products of other mutations, such as losses or gains of other gene functions. Structural variation is common in the LTEE populations, with many more large deletions than duplications (92). Gene duplications can give rise to robustness because they provide functional redundancy (55), whereas deletions might increase the essentiality of other genes by eliminating some redundancies. Because we observed more genes becoming essential than vice versa, we examined whether changes in essentiality might be associated with the presence or absence of redundant genes.

To that end, we sequenced the ancestors and evolved strains at 50,000 generations with high coverage ($>60\times$) and identified all large deletions in the evolved genomes. We identified 251 groups of homologs with $>40\%$ identity in the ancestral genome (Data S3). In each evolved strain, between 8 and 36 homolog groups had been reduced to a single surviving member after 50,000 generations.

We therefore asked if the newly essential genes could be explained by the potential loss of functional redundancy caused by these gene deletions. Most evolved strains (all except clones from Ara+5 and Ara+6) have deletions in the *manB-cpsG* region, which spans the *rfb* operon. In the strains with these deletions, insertions in *rffG* and *rffH*, which are paralogs of *rfbA* and *rfbB*, were highly deleterious. The insertion mutants in strains without paralogs had fitness indistinguishable from lethality, whereas the same insertion mutants in strains with the paralogs still present had

stable frequencies (Fig. S13A). These results are consistent with a previous finding that the absence of both genes leads to envelope stress owing to accumulation of a toxic intermediate, the ECA Lipid II (93). In further agreement, we found that deletion of *wecA*, a gene upstream of ECA II biosynthesis, was tolerated in all strains, indicating that gene is nonessential (Data S4).

In another example of gene essentially depending on functional redundancy, a highly expressed copy of *kdsB* became essential owing to either the loss or very low expression of a duplicate copy. The gene *kdsB* is essential in *E. coli* strain K-12. However, there are two copies of *kdsB* in the LTEE ancestor (ANC, REL606), and both copies have high expression levels (54), making each copy dispensable. In three evolved populations (Ara-3, Ara+1, Ara+2), one copy of *kdsB* (indicated as *kdsB_2* in Data S5) has been lost, making the other copy (*kdsB_1*) essential. In strains with two copies of *kdsB*, whether *kdsB_1* is essential or not depends on the expression level of *kdsB_2*. The fitness effect of insertions in *kdsB_1* is correlated with the expression level of *kdsB_2* (Fig. S25); in strains with high expression levels of *kdsB_2*, the effect of disrupting *kdsB_1* is less deleterious, and vice versa. This example shows that the presence of a duplicate gene is not sufficient to make a gene dispensable; instead, the duplicate must also be expressed at a high enough level. Overall, such examples were uncommon; for the vast majority of newly essential genes (73 of 77) we found no evidence that essentiality was caused by the loss of redundant genes.

Conversely, we asked whether some genes became nonessential owing to duplication of the region containing them. We first focused on the Ara-6 strain because it had an exceptionally high number of genes that had become nonessential by 50,000 generations. We observed a large duplication in the genome of Ara-6 encompassing ~300 genes and a smaller duplication spanning ~25 genes (Fig. S13B). Thirty-eight of the 67 genes that became nonessential in this strain were in those duplicated regions. Three other strains (Ara-3, Ara+2, and Ara+5) had duplicated regions, but only Ara+2 had genes that became nonessential owing to the evolved redundancy (Fig. S13C). Moreover, many genes became nonessential in these and other strains without such duplications. Overall, 52 of 97 genes evolving to become nonessential (~9% of the ancestral essential genes) were not linked to gene duplications. Thus, structural variation accounts for a little under half the loss of essentiality observed.

Restricting our analysis to changes in gene essentiality not explained by structural variation, we still observed significant parallelism across the independent lineages in genes becoming both essential and nonessential (Fig. S13D). These parallel changes in gene essentiality presumably reflect changes in physiological processes that became either more or less critical for fitness in the LTEE environment.

Some of the consistent changes in essentiality we observed are concordant with prior work on physiological changes in the LTEE populations. In particular, we observed changed essentiality in the Entner-Doudoroff (ED) pathway and in cytokinesis. Metabolic analysis of the LTEE strains identified significantly increased flux through the ED pathway in four evolved strains (Ara+1, Ara+2, Ara+3, Ara-4) (94). In this study, we found that disruption of *eda*, an enzyme in the ED pathway, had highly deleterious effects in those strains, suggesting the ED pathway plays an important role in increased fitness in the LTEE. Cell size has increased substantially over the LTEE (95), potentially reducing the nucleoid/volume ratio. Our results show that genes involved in cytokinesis (*minD*, *minE*, *dedD*, *ftsP*) became more essential in the evolved populations. We

hypothesize that, as cells become larger, disrupting the min system became more deleterious as the nucleoid occupies relatively less volume. As a consequence, the nucleoid occlusion system may become less effective at preventing aberrant Z-ring formation, thereby increasing the frequency of minicell formation and reducing fitness. More generally, we hypothesize that many other as yet unexplained changes in gene essentiality are similarly the indirect result of selection on various metabolic pathways.

Supplementary Figures



Fig. S1: Beyond strict loss-of-function effects. Examples of important sub-genic structure for known targets of selection in the LTEE (*mrdA*, *nadR*, *spoT* and *topA*). We detected beneficial insertions mapping to preceding regions of the same transcription unit (*mrdA*, *nadR*, and *topA*), suggesting alterations at the transcriptional level. More strikingly, we observed beneficial insertions mapping to the C-terminal portion of essential genes (*serB*, *topA*, and *spoT*), which presumably cause only partial loss of (or even alterations in) functionality.

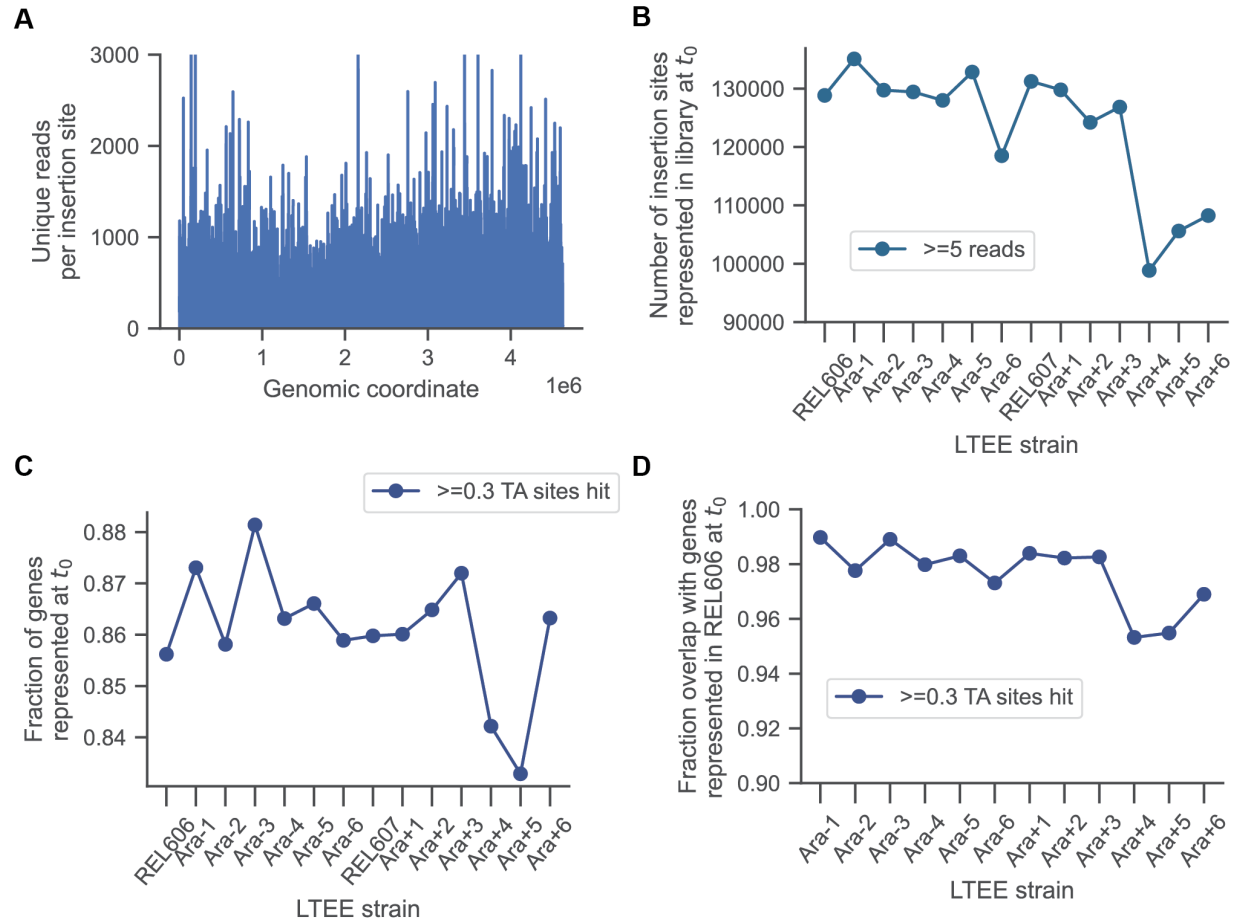


Fig. S2: Distribution of transposon insertions and representation of genes in the transposon sequencing dataset. (A) Insertions in the ancestor REL606 (a representative example) are distributed throughout the genome. (B) Number of insertions at the timepoint zero with at least five mapped reads (Note that the total number of available insertions in the ancestor is 211,995). (C) Fraction of genes represented (at least 30% of the interior TA sites with 5 or more mapped reads) at the timepoint zero. (D) Fraction of genes represented in the ancestor, REL606, that are also represented in the evolved strains.

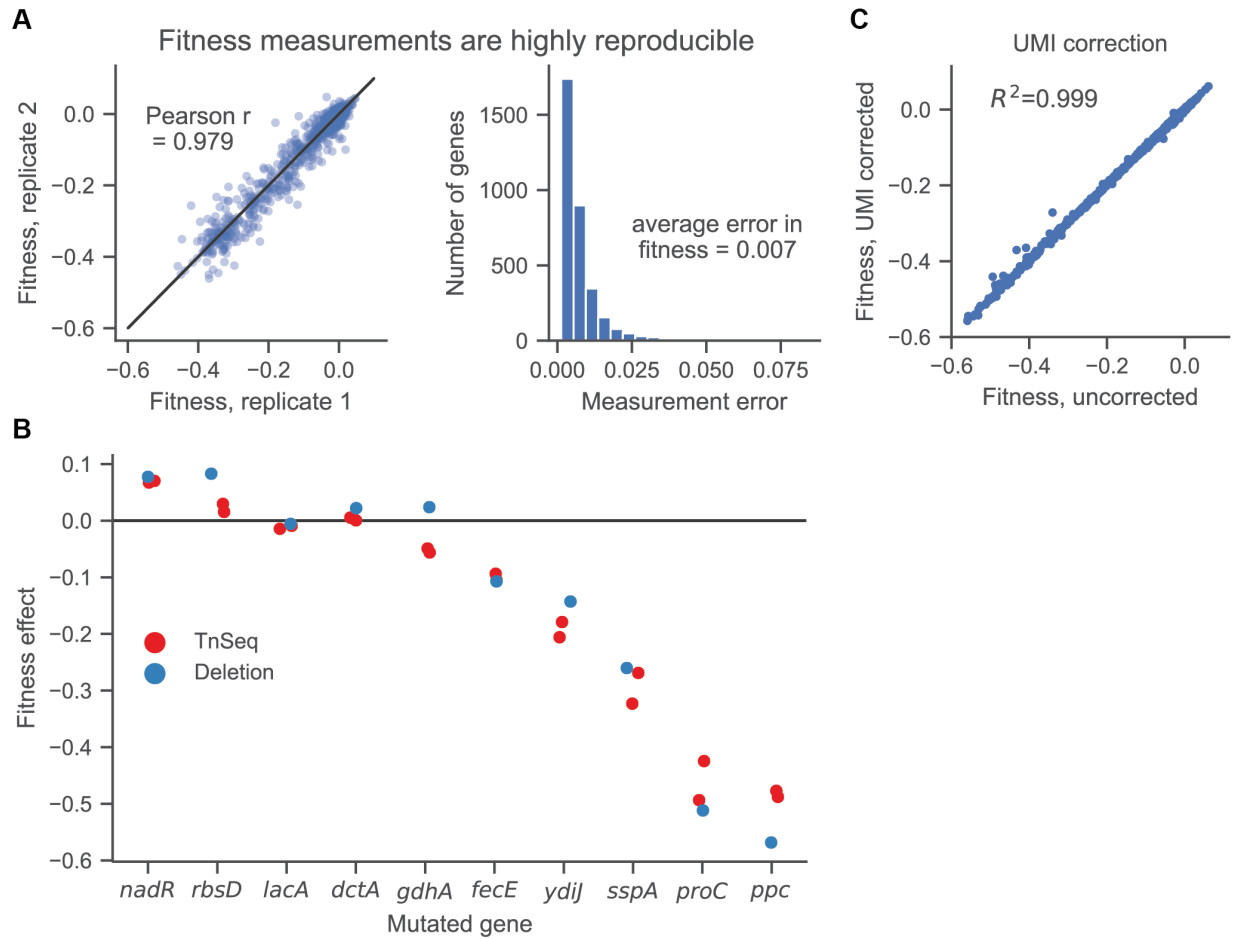


Fig. S3: Estimates of fitness effects of TnSeq insertion mutations are highly reproducible and consistent with fitness effects of deletions in the same genes. (A) Correlation between technical replicates in evolved Ara+1 clone (left) and distribution of measurement errors (right). (B) The fitness effects estimated from disrupting genes using TnSeq in the LTEE ancestor and by pairwise competitions between the unmutated ancestor and clean deletion mutations are consistent. Each point represents an independent measurement. (C) Fitness estimates with and without UMI correction are very consistent with each other.

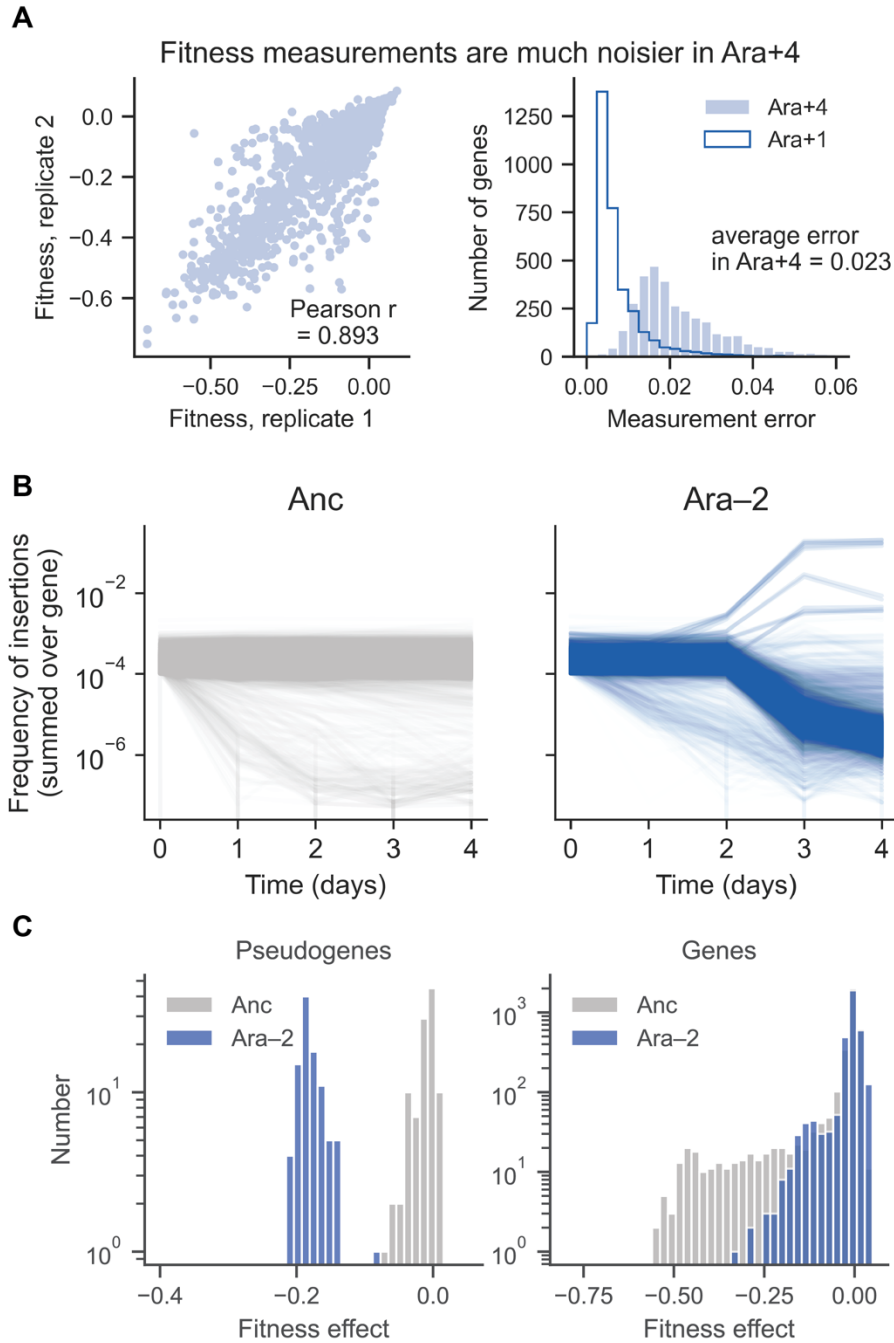


Fig. S4: Evolved strains from populations Ara-2 and Ara+4 were excluded from further analyses because of aberrant properties. (A) The correlation between technical replicates is much worse for Ara+4, and the typical measurement error much larger (2.3%), than for other strains (ranges from 0.7 to 1.2%). **(B)** Plot of all mutant trajectories in the LTEE Ancestor and Ara-2. **(C)** Fitness estimates are biased in Ara-2 because of the highly negative correction factors based on the pseudogenes (left panel). As a consequence, the DFE for Ara-2 barely extends below -0.3 (right panel).

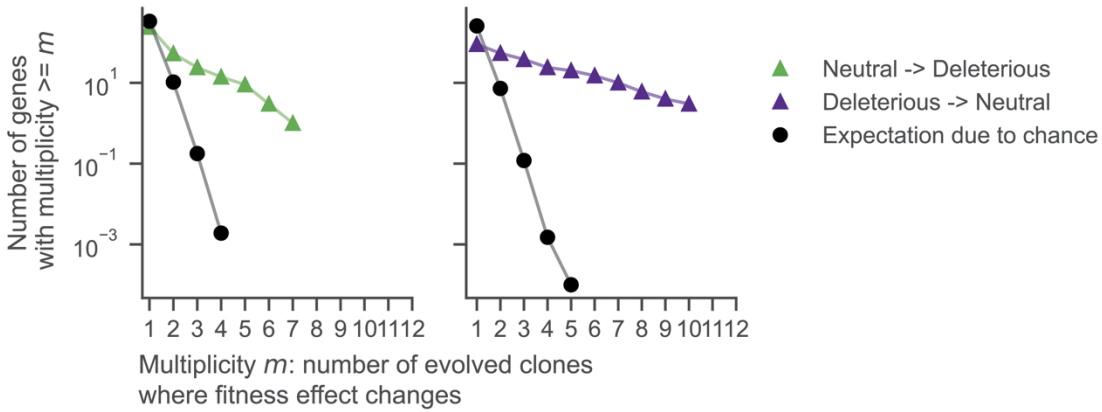
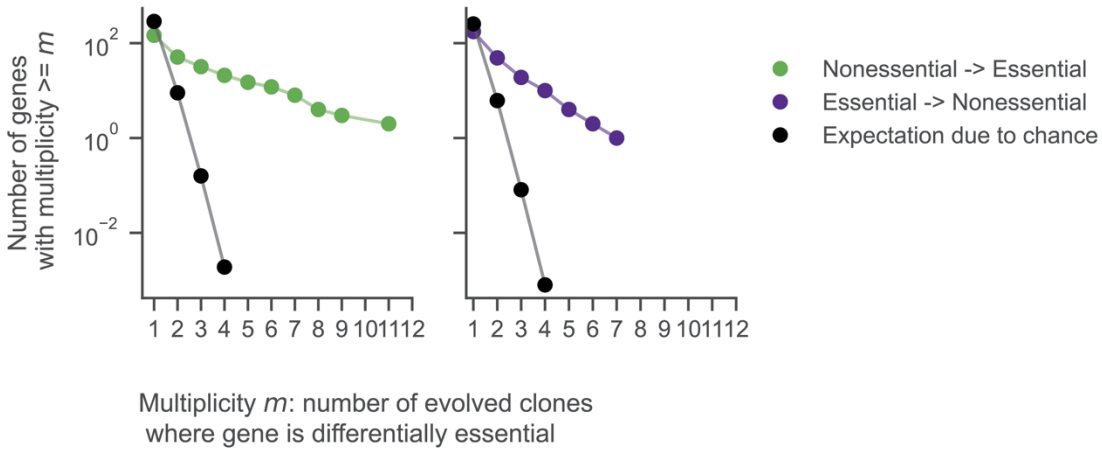
A**B**

Fig. S5: Key observations of parallelism in changing fitness effects and gene essentiality are unaffected when we include Ara-2 and Ara+4 in analysis. (A) Parallel changes in fitness effects. (B) Parallel changes in gene essentiality. We estimated the expected number of parallel changes from chance alone by shuffling the profile of changes in fitness effects 10,000 times and counting how often the same genes had parallel changes (neutral to deleterious or deleterious to neutral) in at least m populations. The expectation is an average over 10,000 simulations, and therefore it can be < 1 .

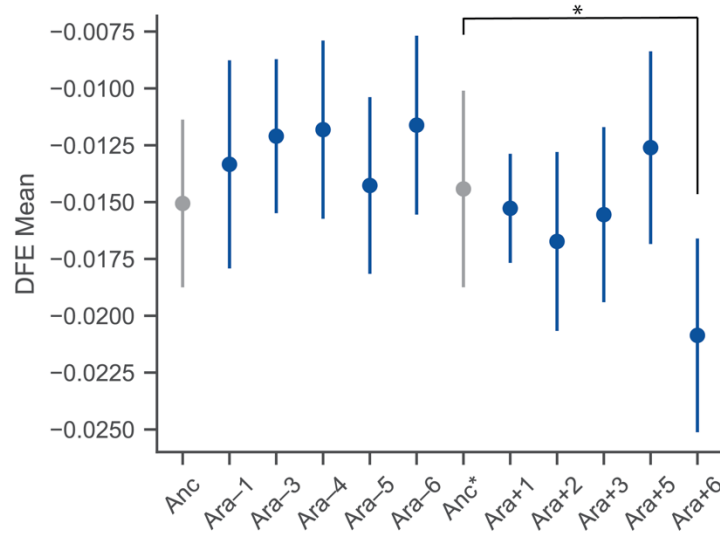


Fig. S6: Differences in the DFE are largely driven by the deleterious tail of fitness effects (Z -test, * $0.01 < p < 0.05$). Here, we compared the means of the DFE for the populations excluding mutations with a fitness effect $s < -0.3$, as these measurements are noisier, and prone to biases due to differences in number of surviving cells, and therefore generations of growth after recovery from the freezer stock.

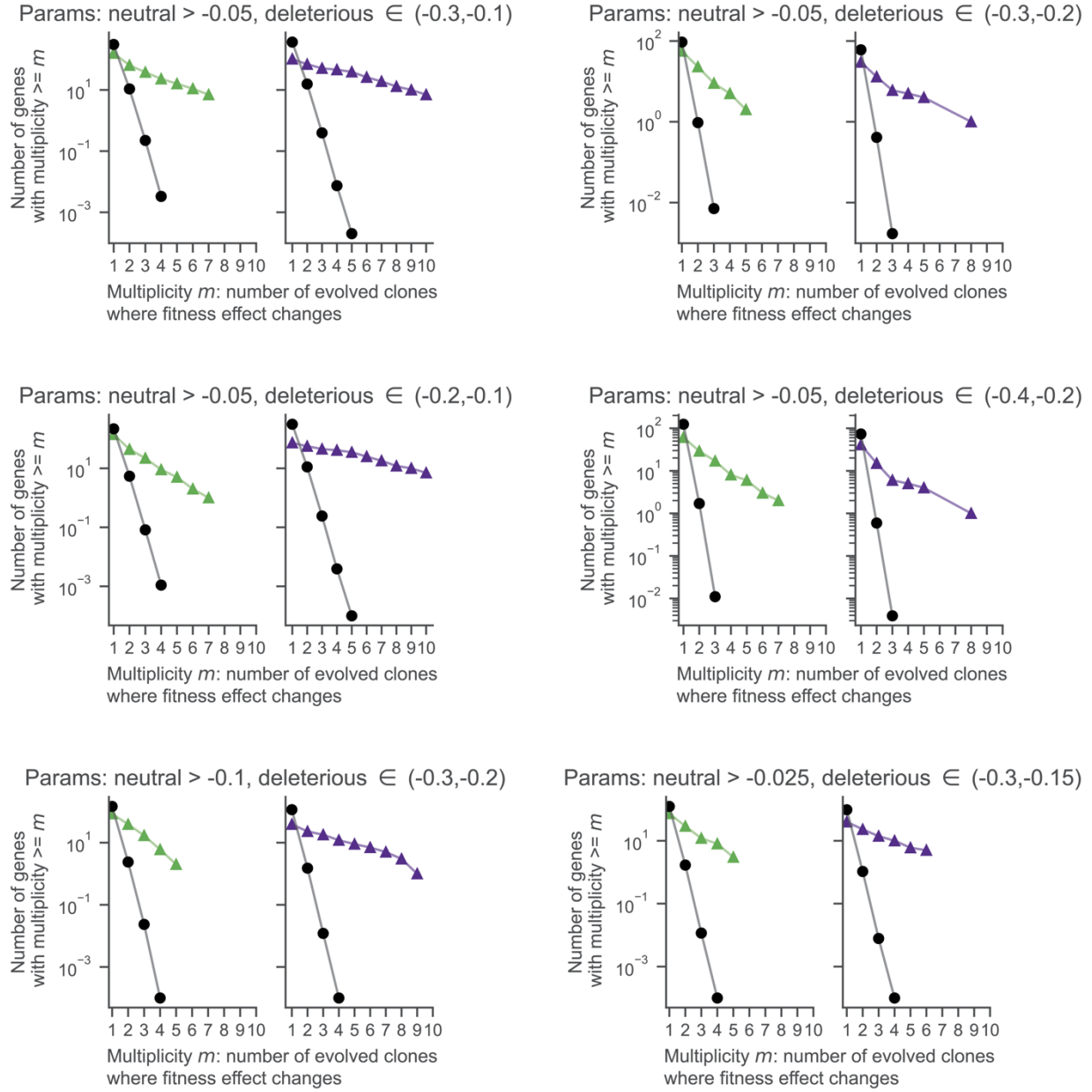


Fig. S7: Signature of parallelism in fitness effect changes is robust to the choice of thresholds for defining neutral and deleterious fitness effects. The number of genes with varying fitness vastly exceeds what would be expected from pure randomness in all these plots. The expectation is an average over 10,000 simulations, and therefore it can be < 1.

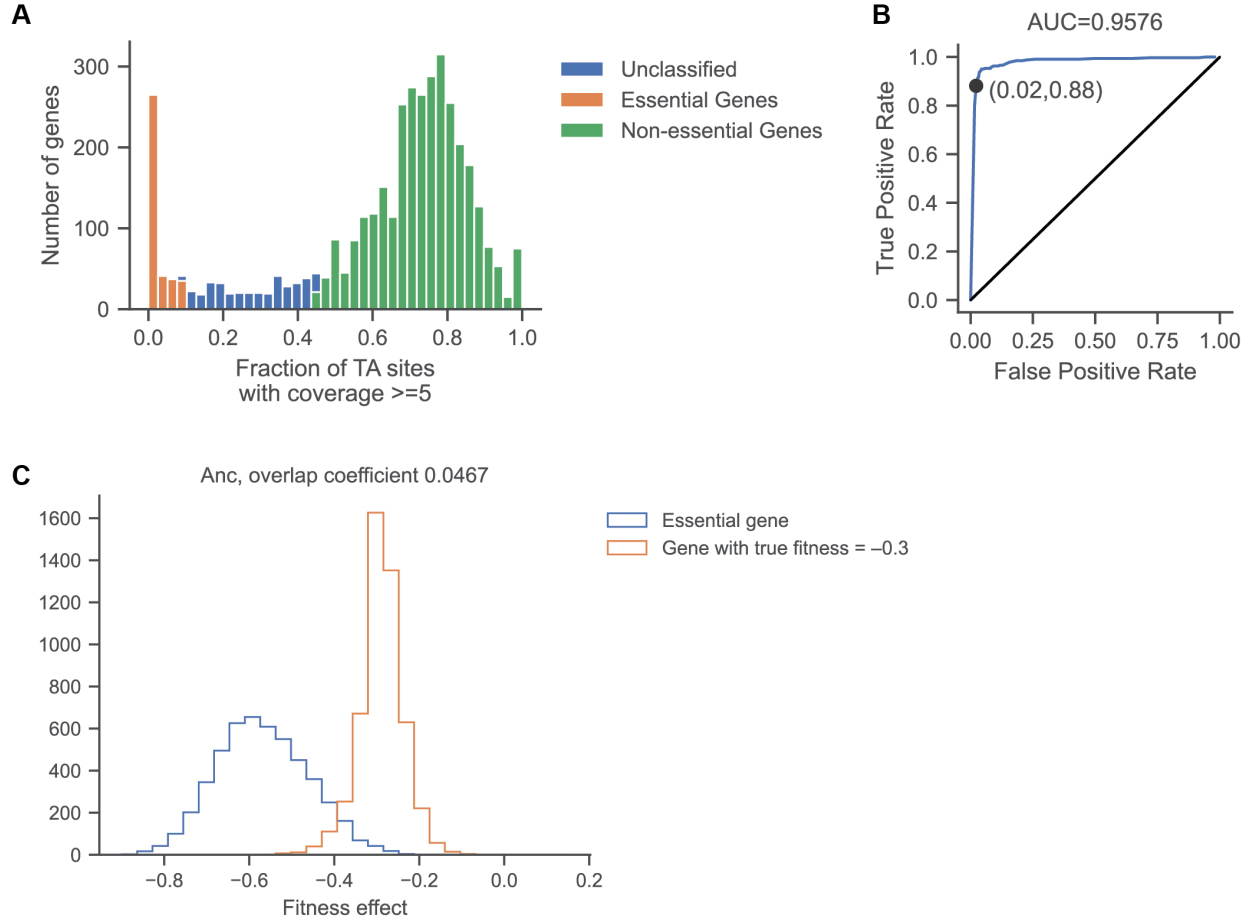


Fig. S8: Defining gene essentiality in LB and DM25. (A) LB: Distribution of fraction of sites with coverage ≥ 5 at t_0 in the fitness assay in the ancestor, REL606. Genes with that fraction < 0.1 are classified as essential and genes with fraction > 0.45 as nonessential. (B) Classification of gene essentiality in LB is reliable when compared to data from Goodall *et al.* (48). (C) DM25: We simulated the fitness estimates for a truly essential gene (no growth during the assay) and for genes with true fitness effects ranging from -0.45 to -0.2 . We found the most deleterious effect, $s = -0.3$, where the overlap coefficient is < 0.05 . This overlap represents the probability of observing the same fitness effect in a truly essential gene and in a gene disruption that is highly deleterious, and therefore the limit of our fitness assay to distinguish essentiality from significant growth deficit. See Methods: *Data Analysis: Fitness and Essentiality* for a detailed description of these simulations.

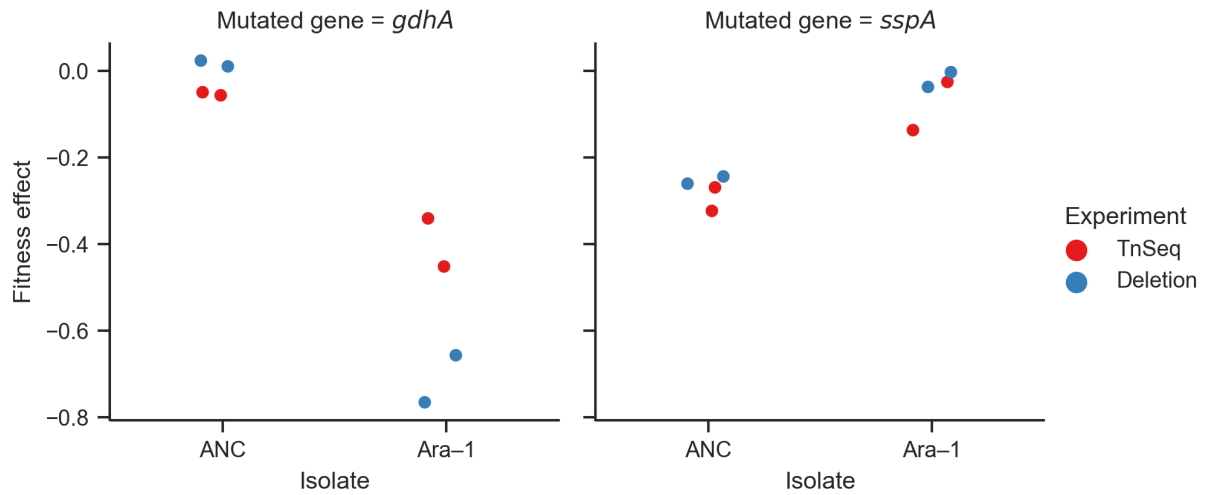


Fig. S9: Verification of differential gene essentiality for two genes (*gdhA* and *sspA*) in the ancestor and an evolved clone from population Ara-1, obtained by generating clean gene deletion mutants in both genetic backgrounds. For both genes, the prediction of differential essentiality from TnSeq is consistent with competition assays using the deletion mutants. For *gdhA* in the Ara-1 background, both methods classify the gene as essential ($s < -0.3$). The difference between the two methods can be large for essential genes, because the measurement errors for both methods are very large as a consequence of the rapid disappearance of unfit mutants that results in extremely low counts.

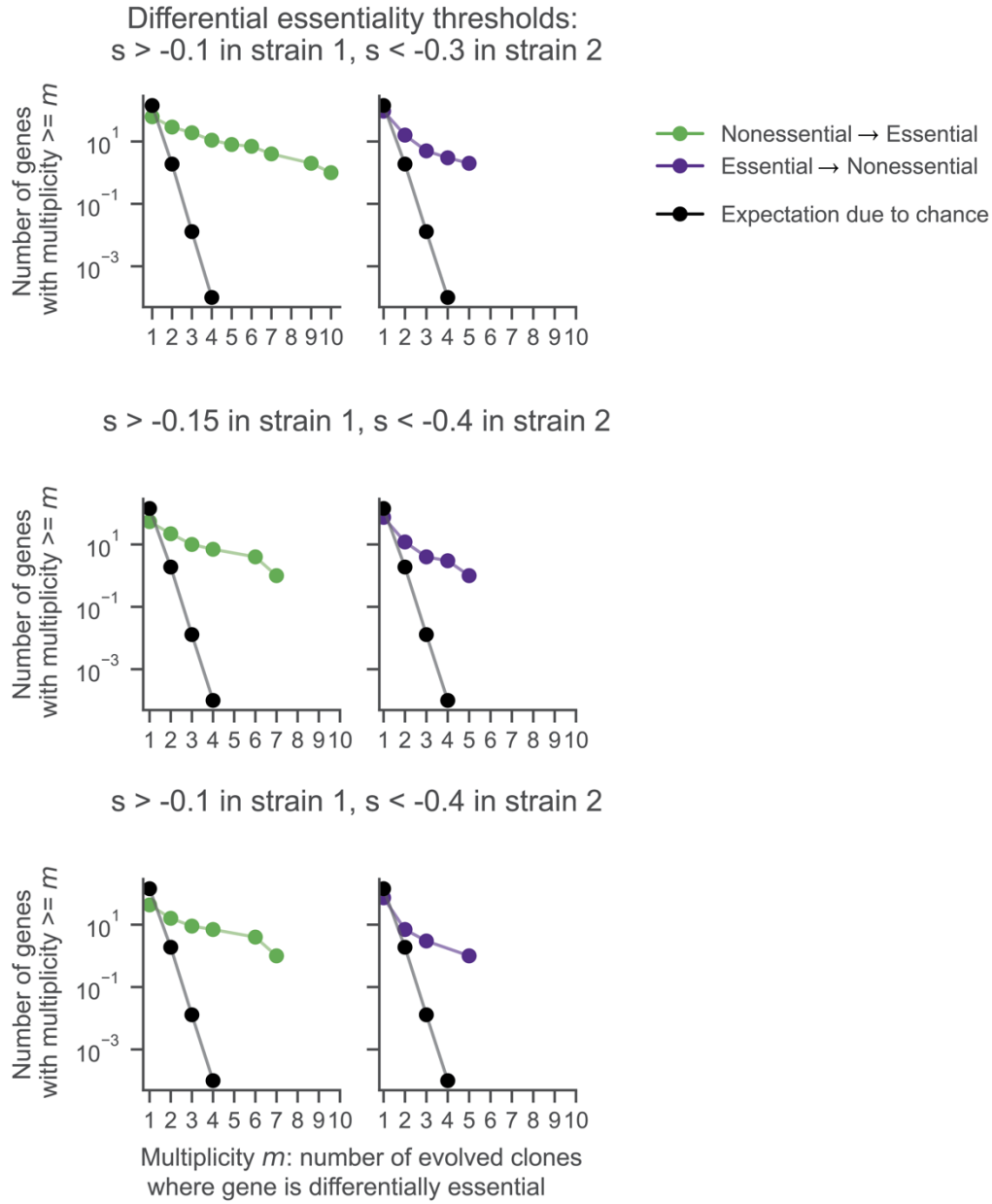


Fig. S10: Signature of parallelism in changing gene essentiality is robust to the choice of thresholds for defining differential essentiality. The number of genes with changed essentiality vastly exceeds what would be expected from pure randomness in all these plots. The expectation is an average over 10,000 simulations, and therefore it can be < 1 .

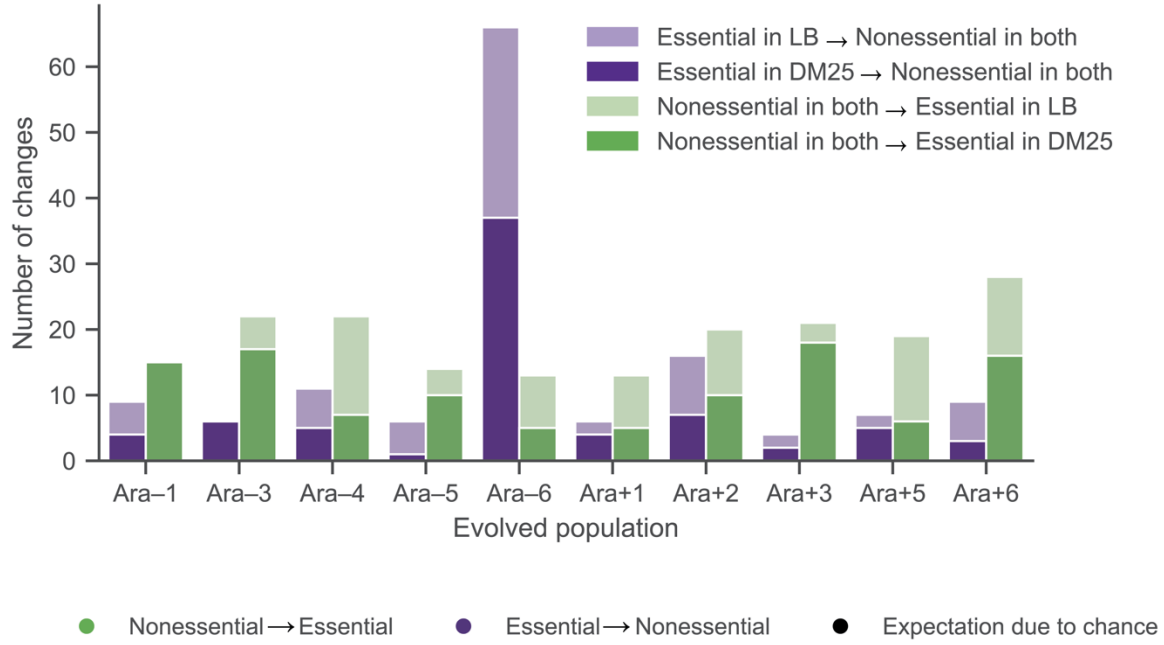
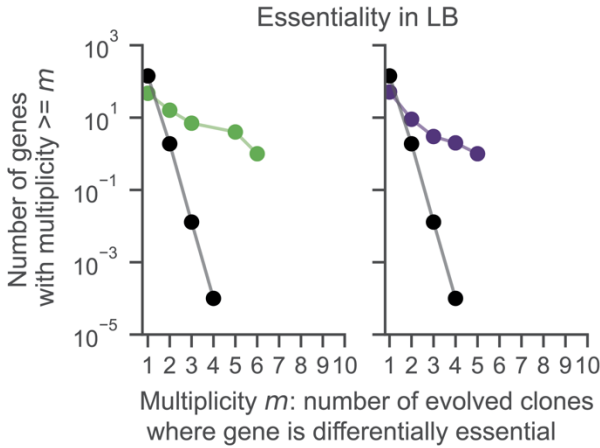
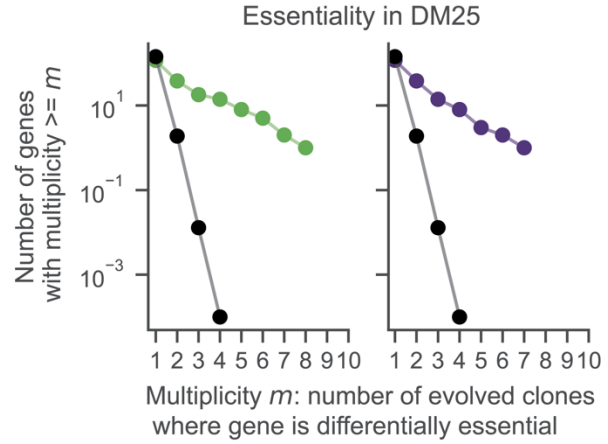
A**B****C**

Fig. S11: Parallelism in gene essentiality changes, partitioned by two different media. (A) Partitioning changes in essentiality by growth medium. Parallel changes in gene essentiality in (B) LB and (C) DM25 media. We estimated the expected number of parallel changes from chance alone by shuffling the profiles of changes in gene essentiality 10,000 times and counting how often the same genes had altered essentially in at least m populations. The expectation is an average over 10,000 simulations, and therefore it can be < 1 .

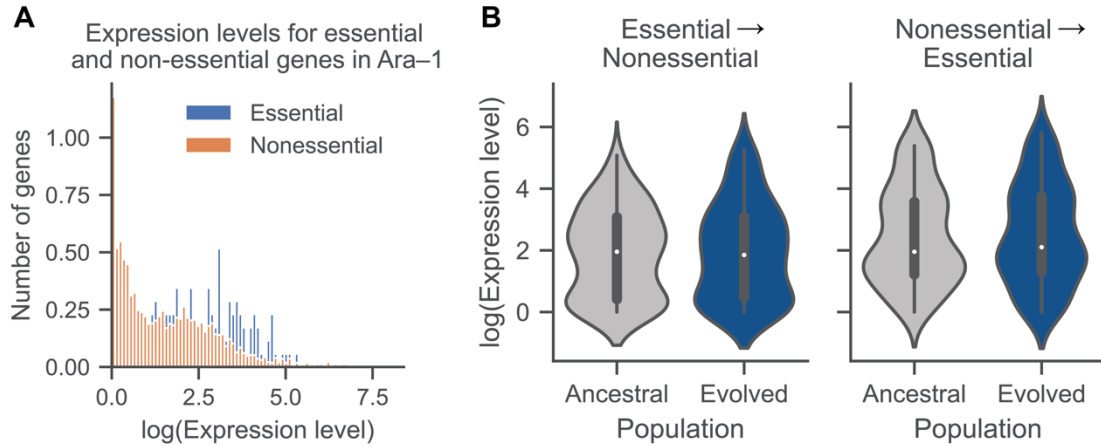


Fig. S12: Changes in gene essentiality in DM25 are not related to differences in expression levels. (A) Nonessential genes tend to have lower expression levels. The pattern shown for Ara-1 occurs in all LTEE clones (see Jupyter Notebook `generate_figures_main.ipynb` section on expression levels comparison). (B) Changes in gene essentiality do not reflect systematic differences in expression levels between the ancestor and evolved strains. Here, we show all genes with altered essentiality in any evolved strain. We compare the expression level of the gene in the ancestor and the evolved strain with altered essentiality for all these genes. The baseline expression levels are generally higher in the evolved strains; when comparing strains, we therefore normalized the expression levels of each gene by the total number of reads that mapped to the strain.

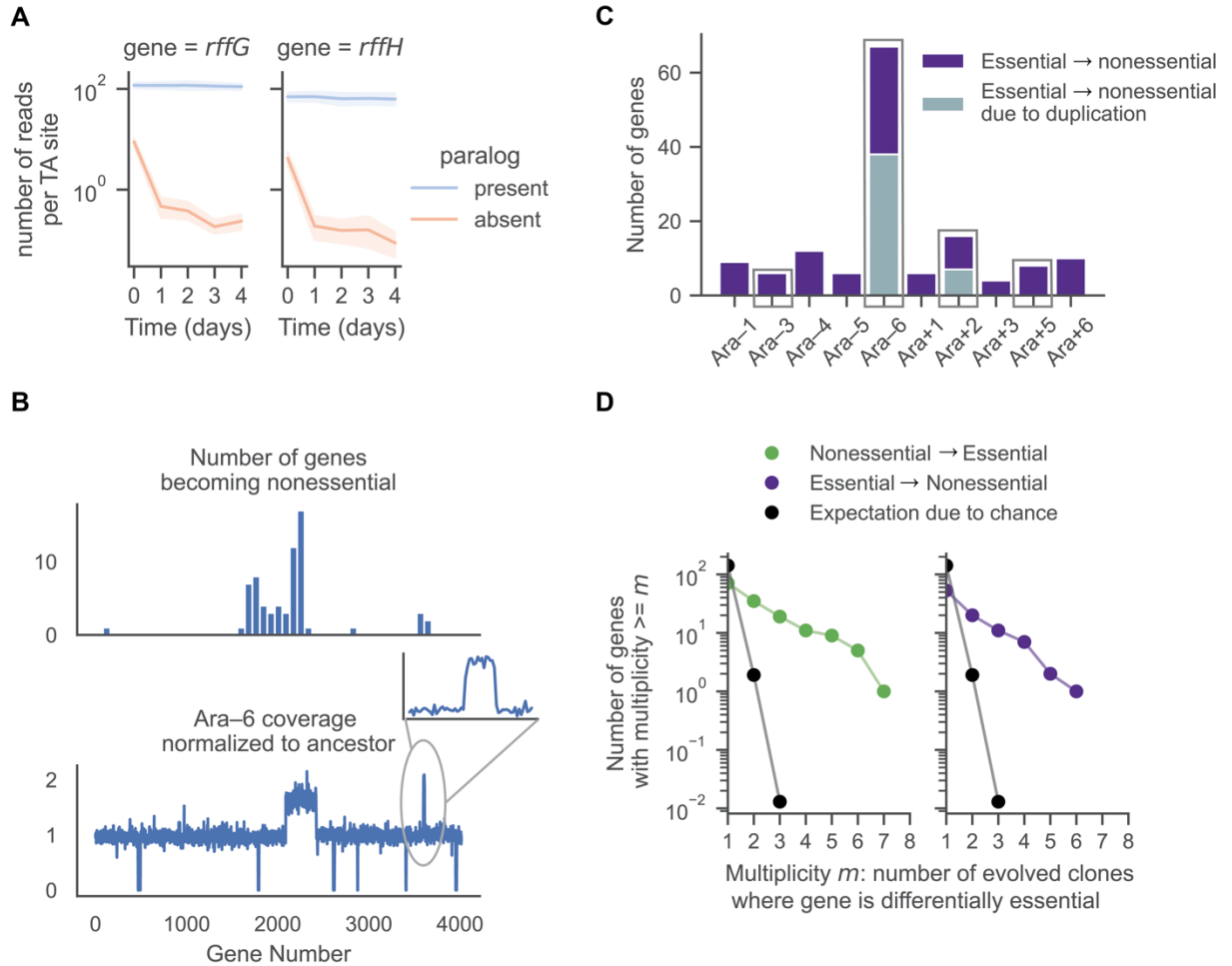


Fig. S13: Gain or loss of functional redundancy explains some, but not most, changes in essentiality. (A) Loss of paralogs makes *rffG* and *rffH* essential. In strains with *rfbA* and *rfbB* present (ancestors, Ara+5, Ara+6), insertion mutations in *rffG* and *rffH* are nearly neutral, and they persist over the 4-day bulk fitness assay (~26 generations). In the 8 evolved strains with deletions of the *manB-cpsG* region, insertion mutations in *rffG* and *rffH* are rapidly outcompeted. Note the logarithmic scale for the abundance; <1 read per TA insertion site is at the level of measurement noise. (B) In the 50,000-generation clone from population Ara-6, much of the distribution along the chromosome of the genes that became essential during evolution (top subpanel) maps closely to normalized whole-genome sequencing coverage (bottom subpanel); the inset expands the smaller of two multi-gene duplications. (C) Numbers of essential to nonessential changes in the 10 evolved strains. Gray and blue colors represent the numbers that are attributable to gene duplications and those that are not, respectively. Evolved clones with detectable duplications are outlined by a box. (D) Parallel changes in gene essentiality, after excluding those changes attributable to the loss or gain of redundancy. We estimated the expected number of parallel changes from chance alone by shuffling the gene essentiality and fitness profiles 10,000 times and counting how often the same genes had altered essentially in at least m populations. The expectation is an average over 10,000 simulations, and therefore it can be < 1 .

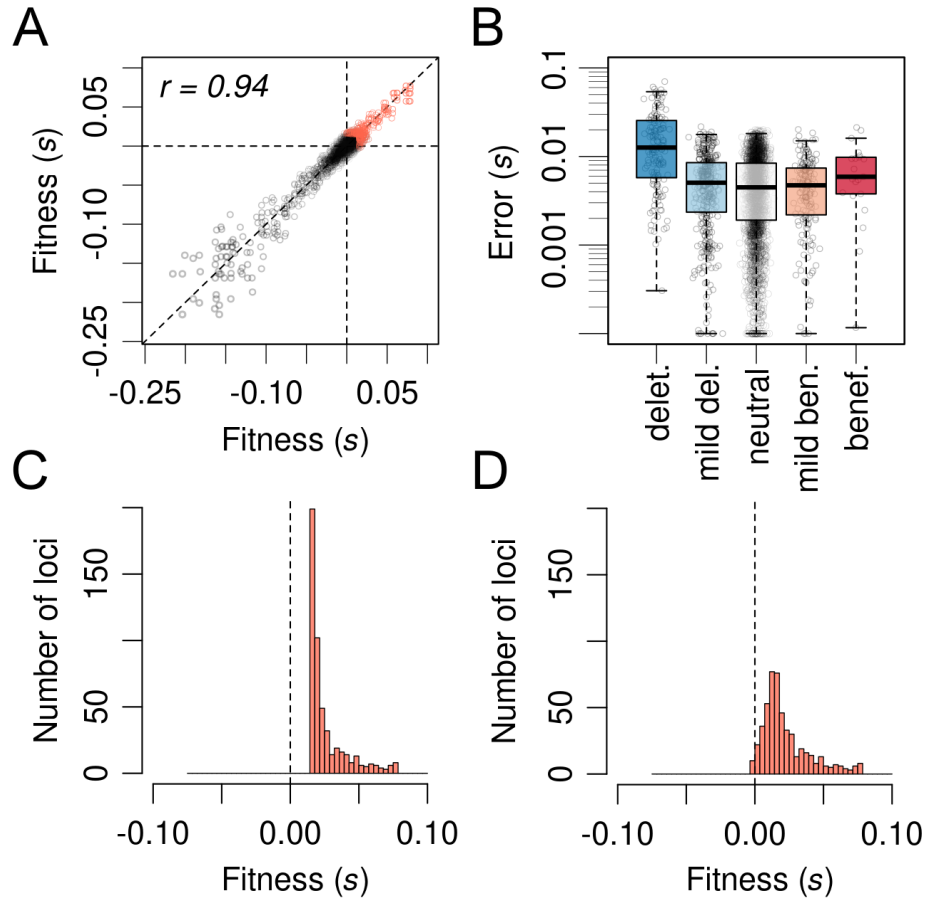


Fig. S14. Reproducibility of fitness estimates (A) Pairwise comparisons among segments from the same locus show high reproducibility. Pearson's coefficient, r , is in upper left corner. (B) Measurement error among pairwise replicates scales with the magnitude of fitness effects (deleterious and beneficial: $|s| > 0.05$; mild: $0.05 > |s| > 0.015$; neutral: $|s| < 0.015$). (C, D) Despite measurement noise, the shape of the DFE for beneficial mutations (C) retains a marked shift towards positive values when constructed using different segments from the same locus (D).

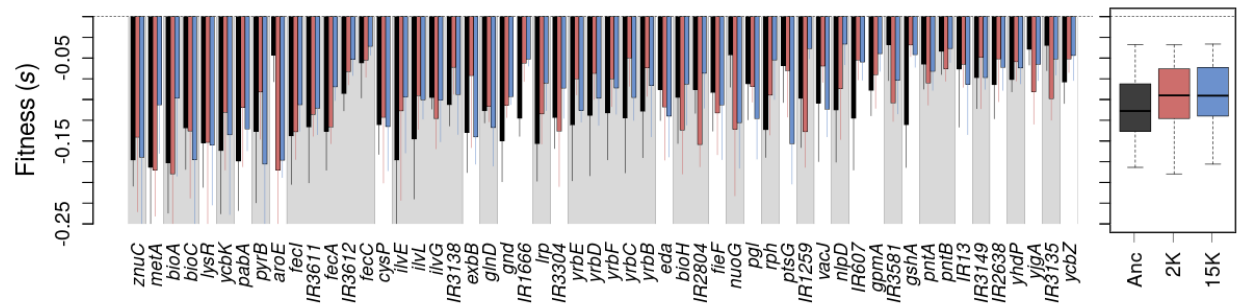


Fig. S15. Effect of genetic background on the fitness effects of mutations in specific gene targets. The genes and intergenic regions subject to the most severely deleterious mutants in three backgrounds in the Ara+2 lineage. Colors indicate ancestor (black), 2K (red), and 15K (blue) evolved strains. Gray shaded areas indicate loci in the same transcription unit. Values show the average across the different segments of each locus. Error bars indicate 95% confidence intervals. Note that many of these loci correspond to genes previously reported as essential for growth in minimal medium (e.g., *metA*, *pyrB*, *aroE*). However, the mutants that we recovered carry insertions only in their C-terminal portion, which presumably cause only partial loss of functionality.

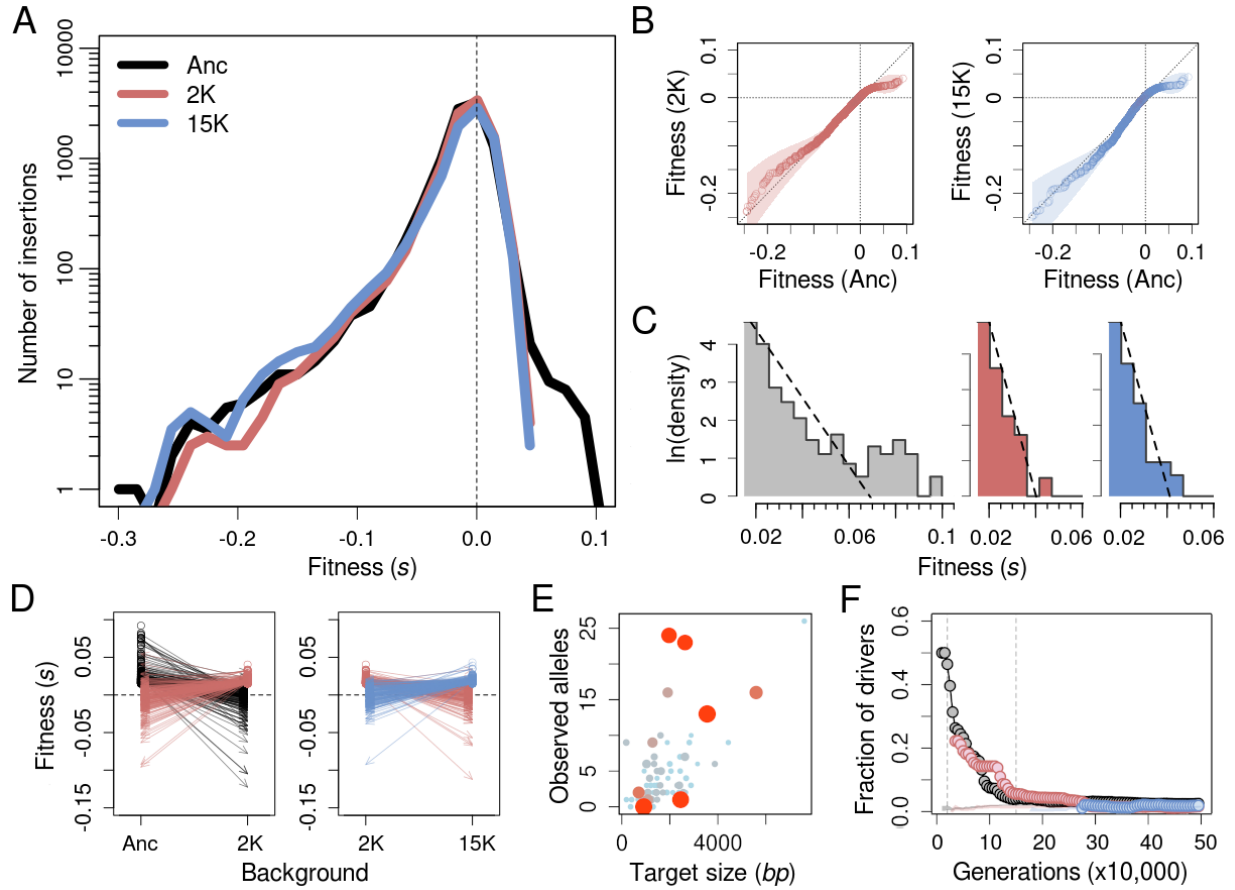


Fig. S16. Change in the DFE along a second large fitness gradient. (A) DFEs in the ancestor (black), 2K (red) and 15K (blue) evolved strains from population Ara-1. Note the logarithmic scaling of the y-axis. (B) Deleterious tails were unchanged during adaptation, as indicated by comparing the cumulative fitness distributions for the ancestor and the evolved strains (2K, left; 15K, right). Shaded areas show 95% bootstrapped confidence intervals. (C) Beneficial tails were rapidly truncated during the LTEE, and they became exponentially distributed ($P < 0.001$ for Anc versus $P = 0.507$ and $P = 0.695$ for Ara-1 clones 2K and 15K, respectively; one-sample K-S test). Histograms show the best fits to exponential distributions (dashed lines) in the ancestor (gray), 2K (red) and 15K (blue) backgrounds. Note that all three x-axes use the same scale. (D) Most beneficial mutations available to the ancestor become neutral or deleterious in the 2K background (black arrows), while most beneficial mutations available in the 2K background were neutral or deleterious in the ancestor (red arrows). The same general pattern occurs when comparing beneficial mutations in the 2K and 15K backgrounds (right panel). (E) The prevalence of mutations in the LTEE shows a good correlation ($r = 0.62$) with the mutational target size; the area and color of the dots represent fitness. (F) The predictive capacity of the DFEs as a function of time in the LTEE. Values show the fraction of numerically dominant alleles at each generation captured by the DFE measured in the ancestor (black), 2K (red), and 15K (blue) evolved strains. Shaded areas show the null expectations based on randomly sampling neutral and deleterious mutations. The results shown here in all panels (A-F) for LTEE population Ara-1 are strikingly similar to those seen for the independently evolving Ara+2 population (see Fig. 5A-C, E, and Fig. 6A,D).

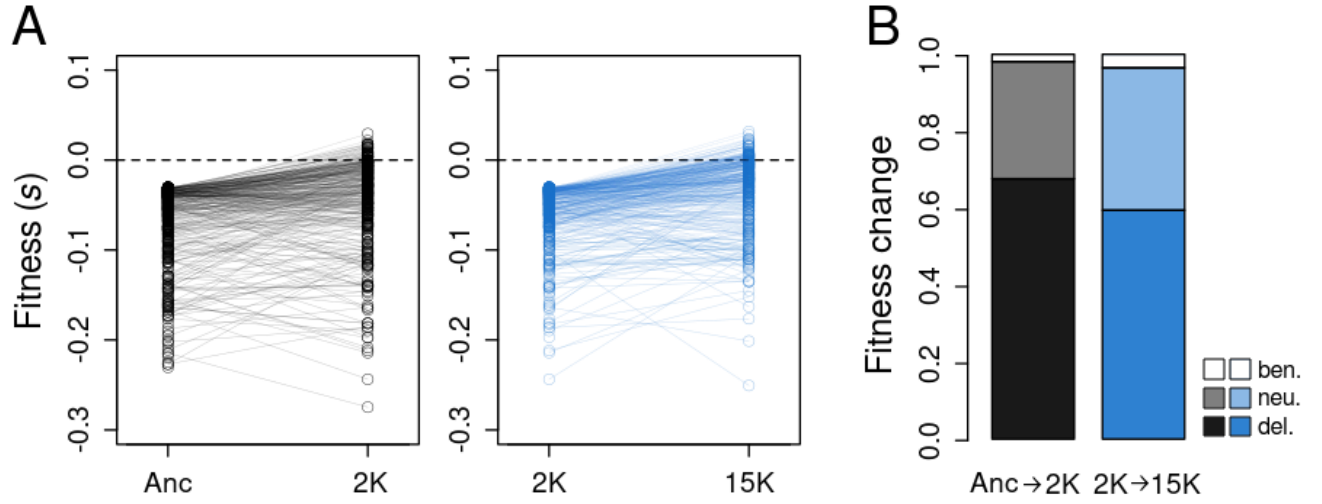


Fig. S17. Patterns of epistasis among deleterious mutations in Ara+2 lineage. (A) Most deleterious mutations in the ancestral background remain deleterious in the evolved 2K background (left), and most deleterious mutations in the 2K background remain deleterious in the 15K background (right). (B) The fraction of deleterious mutations with deleterious, approximately neutral, and beneficial effects in more fit backgrounds. Most changes from deleterious to neutral or beneficial effects occur in mutations with fitness effects close to the detection limit for neutrality in the earlier background ($s = -0.015$).

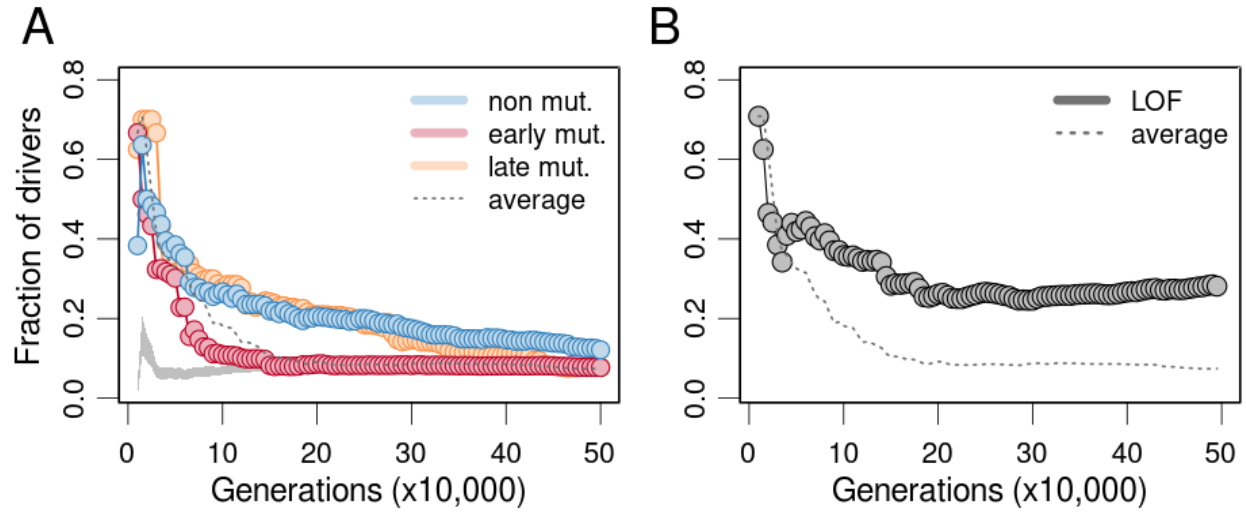


Fig. S18. Loss-of-function mutations and the predictability of adaptation. (A) Predictive capacity of the ancestral DFE as a function of time for different sets of LTEE lineages. Colors indicate analyses performed separately using data from the non-mutator (blue), early mutator (orange), and late mutator lineages (red). The dashed line shows the aggregate average when pooling all lineages, as shown in Fig. 4D. The shaded area shows the null expectation based on randomly sampling neutral and deleterious mutations. (B) The fraction of loci harboring high-frequency beneficial mutations that show mutations with presumed loss-of-function effects (nonsense, frameshift, and structural variants). The fraction of these important targets of adaptation captured by the DFE measured in the ancestor is shown for comparison (dashed line).

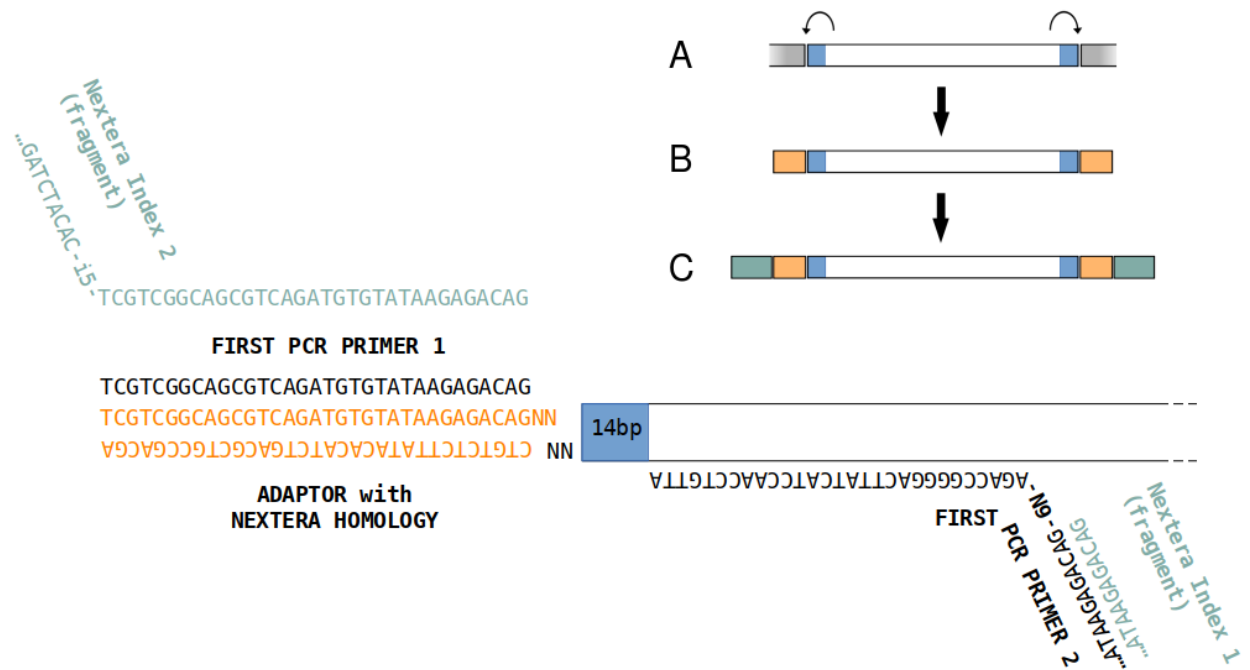


Fig. S19. Procedure for capturing the transposon flanking regions. (A) Genomic DNA is extracted in bulk, digested with *MmeI*, and separated by gel electrophoresis. (B) Transposon-sized fragments are ligated with double-stranded oligonucleotide adaptors. (C) Two rounds of PCR are performed to add indices for Illumina sequencing.

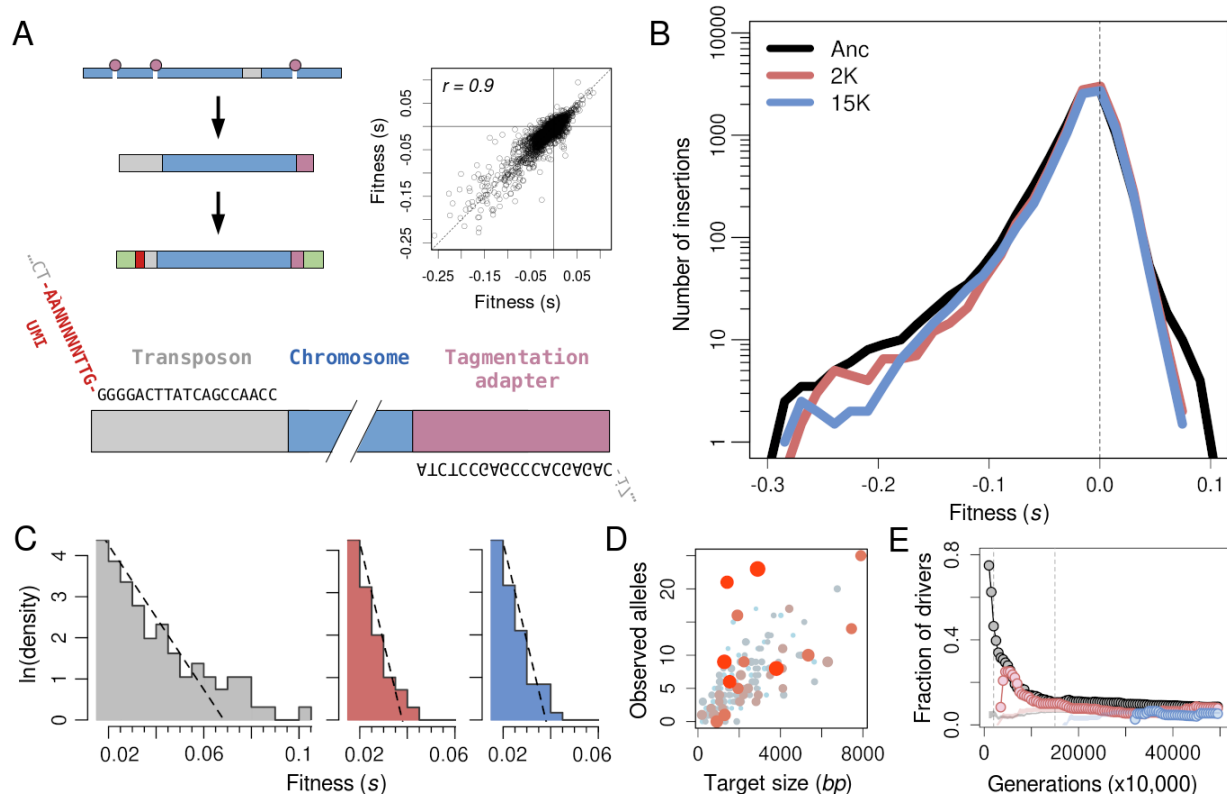


Fig. S20. Additional replicate for Ara+2 samples using the UMI-Seq approach. (A) The libraries generated with the INSeq transposon were tagmented and subjected to two rounds of PCR to add indexes and UMIs for Illumina sequencing. Fitness measurements show good reproducibility between the INSeq and UMI-Seq approach (inset). (B) DFEs in the ancestor (black), 2K (red), and 15K (blue) strains from population Ara+2, showing that the neutral and deleterious fractions of the distributions were largely unchanged during adaptation. Note the logarithmic scaling of the y-axis. (C) Beneficial tails were rapidly truncated and became exponentially distributed ($P < 0.001$ for Anc versus $P = 0.996$ and $P = 0.556$ for Ara+2 clones 2K and 15K, respectively; one-sample K-S test). Dashed lines show the best fit to an exponential distribution in the ancestor (gray), 2K (red), and 15K (blue) backgrounds. (D) The prevalence of mutations in the LTEE shows a good correlation ($r = 0.68$) with the mutational target size; the area and color of the dots represent fitness. (E) The predictive capacity of the DFEs as a function of time in the LTEE. Colors indicate the DFE measured in the ancestor (black), 2K (red), and 15K (blue) strains. Shaded areas below show the null expectations based on randomly sampling neutral and deleterious mutations. The results shown here (panels B-E) recover all the salient features observed in the INSeq replicate discussed in the main text (see Fig. 2A-C, Fig. 3C, and Fig. 4A,D).

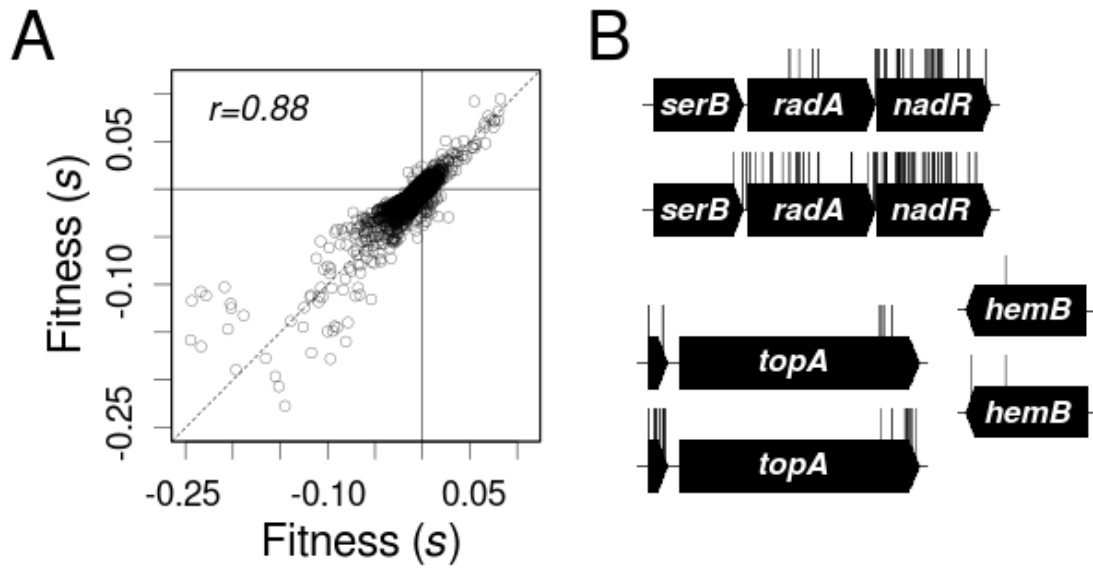


Fig. S21. Comparison of fitness estimates between two datasets. (A) Correlation between selection coefficients measured in the ancestral genetic background in independent datasets for two LTEE lineages (Ara+2 on the x-axis; Ara-1 on the y-axis). Pearson's coefficient, r , is shown in the upper left corner. (B) Lower coverage in the Ara-1 dataset resulted in lower resolution at the sub-genic level. This difference is best illustrated by mapping insertions in essential genes for which insertions in the C-terminal end are neutral (*hemB*) or even beneficial (*serB*, *topA*). The diversity of insertion sites in these loci is consistently lower in Ara-1 (top alignments) than in Ara+2 (bottom alignments).

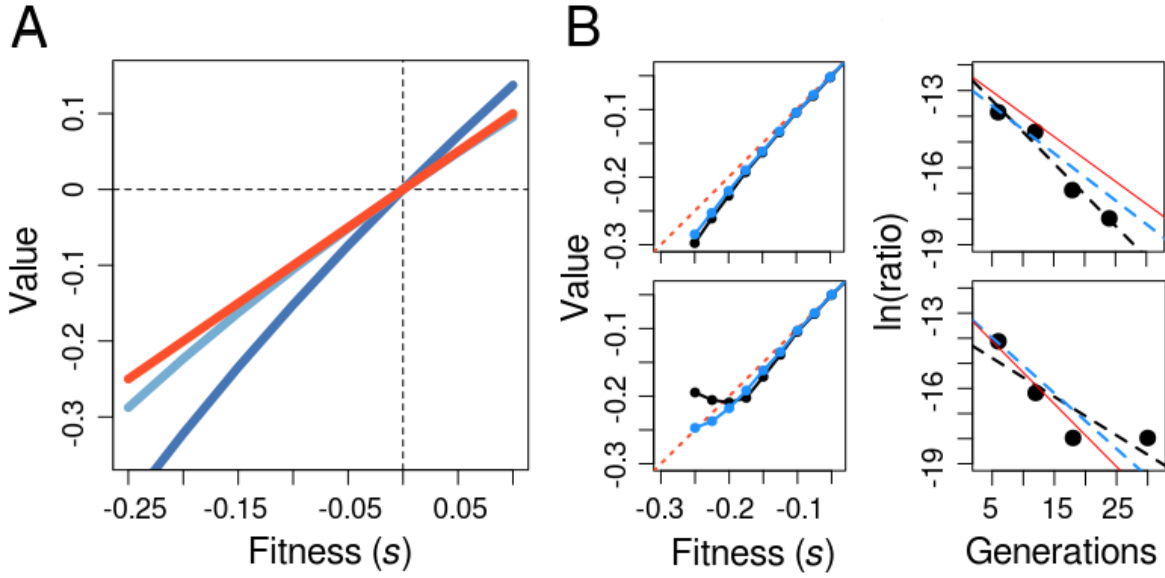


Fig. S22. Measuring fitness effects from bulk competition data. (A) Relationship between the fitness effect (red; often called the selection coefficient, s , in population-genetics literature) and the ratio of realized growth rates (dark blue; typically used as the measure of relative fitness, W , in the LTEE). Empirical estimates of s based on the slope of allele frequency trajectories make the approximation $s \approx \ln(1 + s)$, which equals W multiplied by $\ln(2)$ (light blue) (80). Note, however, that this approximation starts to deviate at large values of s . (B) Weighting the regression by the abundance of each allele improves estimation of s . This improvement is most important at low values of s , when the abundance of a highly deleterious allele approaches zero at later timepoints. Lines show the true s (red), standard regression estimate (black), and weighted regression estimate (blue). Top left panel shows results from a set of simulations with a single deleterious allele starting at a frequency of 7×10^{-7} , and the top right panel shows a single run with $s = -0.18$. Bottom left panel shows results from simulations in which the deleterious allele started at a frequency of 2×10^{-7} , and the bottom right panel shows a single run with $s = -0.25$. Each symbol in the two left panels shows the average of 100 simulations at each value of s . Simulation model adapted from (96), using a total population size of 6×10^7 .

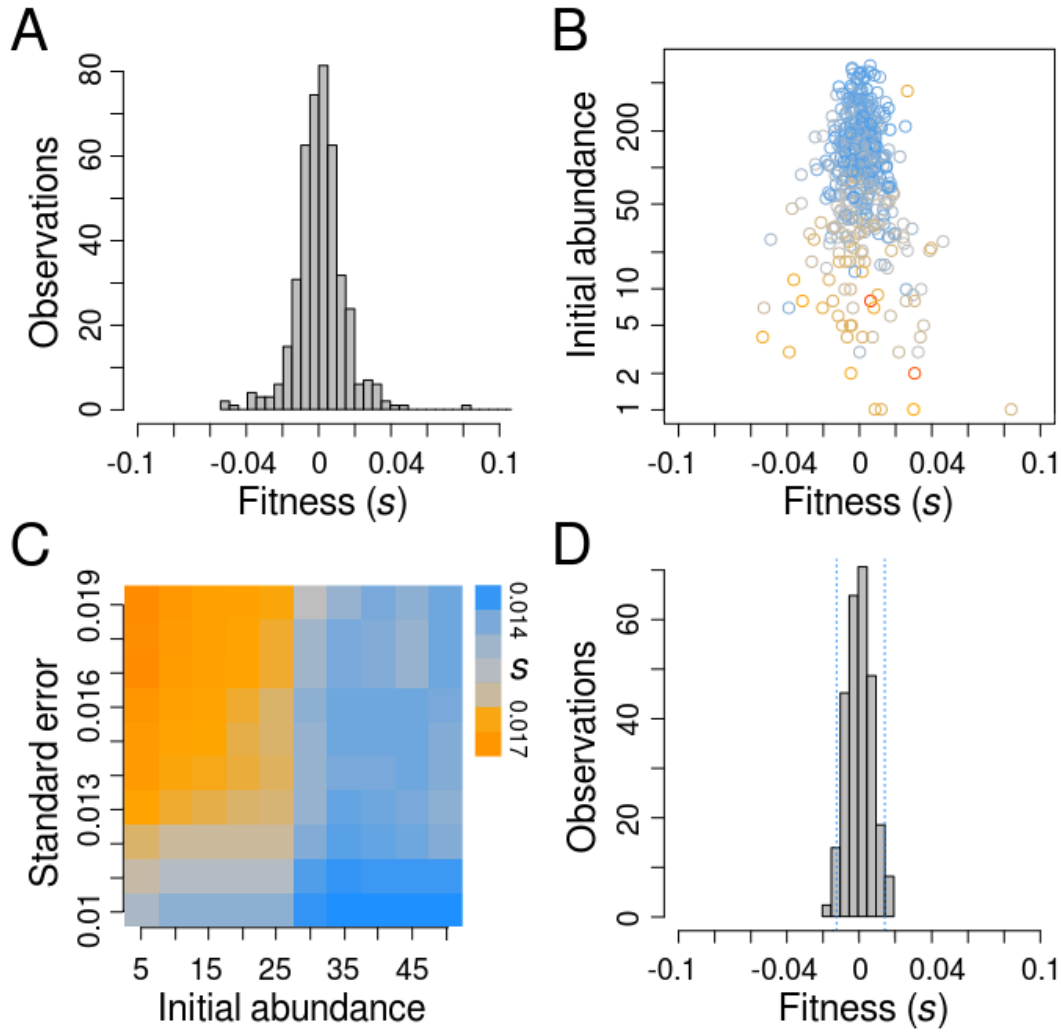


Fig. S23. Data filtering and the threshold of neutrality. (A) Raw distribution of fitness effects for presumed neutral loci in datasets from the ancestor and both evolved Ara+2 strains. (B) Most small deviations from zero can be explained by sampling error, while larger deviations result from low initial abundance (y-axis, note logarithmic scaling) or poor linear fit (colors from blue to gray to orange indicate increasing standard errors). (C) After filtering artifacts, we scanned combinations of initial abundance and standard error to identify selection coefficients encompassing 97% percent of the data (colors, see scale bar on right). (D) Final distribution of selection coefficients for presumed neutral loci after applying all the filters. The blue dotted lines indicate the resulting threshold for approximate neutrality ($-0.015 < s < 0.015$).

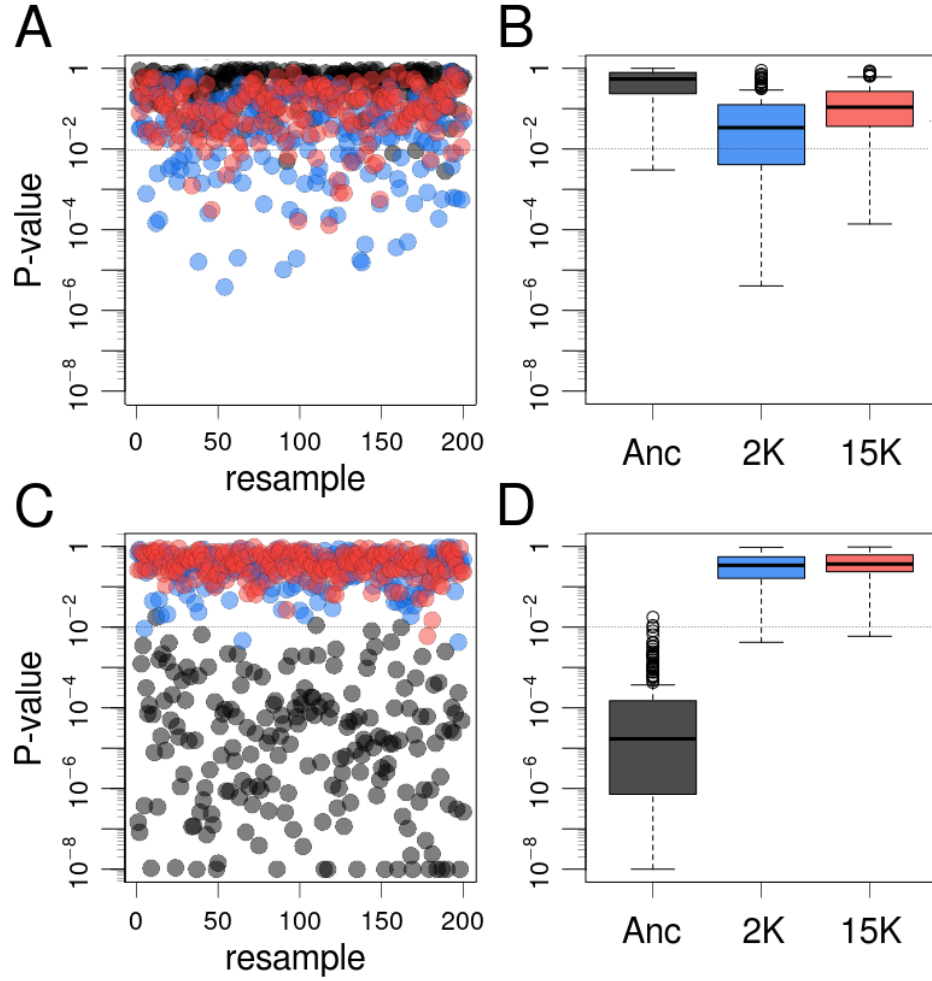


Fig. S24. Measurement error and reliability of tests on the shape of DFEs. To assess the role of noise in statistical analyses of the shape of DFEs, we resampled the distribution tails using regression standard errors as weights. **(A)** P-values from replicated two-sample Kolmogorov–Smirnov (K-S) tests in which the ancestor’s deleterious tail was compared with 200 resampled deleterious tails obtained from the ancestor (used as control, black), 2K (red), and 15K (blue) backgrounds from the Ara+2 lineage. Note the logarithmic scaling of the y-axis. **(B)** Boxplots show that deleterious tails in the more-fit evolved backgrounds are similar to the ancestor’s tail, although for the 2K background the results are ambiguous (average P-values were 0.51, 0.096 and 0.18 respectively). **(C)** P-values from replicated one-sample K-S tests in which an exponential fit was made for 200 resampled deleterious tails from the ancestor (black), 2K (red), and 15K (blue) backgrounds. **(D)** Boxplots show that the beneficial tails in the evolved backgrounds are well fit by an exponential distribution, but this distribution is rejected for the ancestor (average P-values are 1.7×10^{-5} , 0.34, and 0.37 respectively).

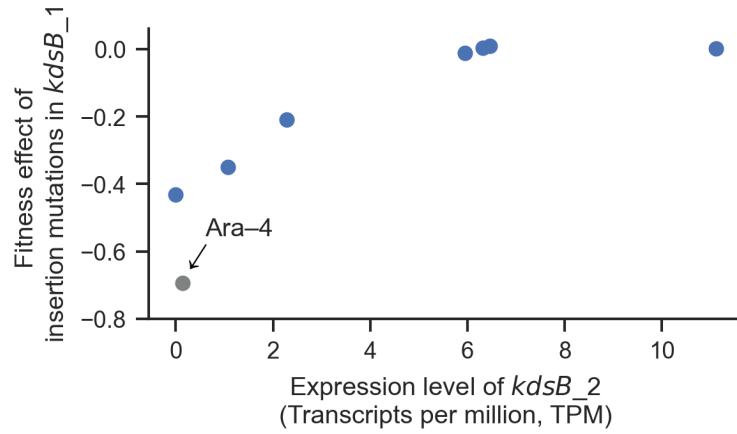


Fig. S25: Fitness effect of disrupting *kdsB_1* gene is correlated with expression (transcripts per million, TPM) of *kdsB_2*. A fitness effect could not be directly calculated for the outlier Ara-4 strain, owing to the mutants' rapid loss from the bulk competition assay. Pearson $r = 0.884$, $p = 0.0083$ excluding Ara-4; Pearson $r = 0.714$, $p = 0.0083$ with Ara-4 set to maximum deleterious effect of $-\ln(100)/\log_2(100) \cong -0.693$.

Supplementary Tables:

Table S1. Accumulated mutations in samples used in the INSeq approach

Name	mutated loci ¹	source
Ancestor	None	(47)
Ara+2 2K	<i>pykF</i> , <i>wcaM</i> , <i>infB</i> , <i>yhgl/gntT</i> , <i>pitA</i> , <i>spoT</i> , <i>rbsD</i> –[<i>rbsB</i>], <i>ECB_03710</i>	(49)
Ara+2 15K	<i>insL-2/lon</i> , <i>ybaL</i> , [<i>nmpC</i>]–[<i>insK-1</i>], <i>lrp</i> , <i>rpsA</i> , <i>appC</i> , <i>flgB</i> , <i>rnb</i> , <i>speG</i> , <i>pykF</i> , [<i>lpp</i>]–[<i>ydiM</i>], [<i>lpp</i>]–[<i>ydiM</i>], [<i>wzy</i>]–[<i>wcaM</i>], <i>wcaM</i> , <i>nupC/yfeA</i> , <i>purL</i> , [<i>serV</i>]– <i>insL-5</i> , <i>yqeB</i> , [<i>ECB_02797</i>]– <i>ECB_02822</i> , <i>mreB</i> , <i>fis/yhdJ</i> , <i>spoT</i> , <i>rbsD</i> –[<i>rbsA</i>], <i>ilvL</i> , <i>yihP</i> , <i>hslU</i> , <i>iclR</i> , <i>metH</i> , <i>sgcR</i> – <i>insA-27</i> , <i>nadR</i> , <i>arcA</i>	(49)
Ara-1 2K	<i>ECB_00736/ECB_00737</i> , <i>topA</i> , <i>pykF</i> , <i>spoT</i> , <i>glmU/atpC</i> , <i>rbsD</i> –[<i>yieO</i>]	(49)
Ara-1 15K	<i>mokC/nhaA</i> , <i>pcnB</i> , <i>insB-2</i> , <i>araJ</i> , <i>ybaL</i> , [<i>nmpC</i>]–[<i>ECB_00513</i>], <i>mrda</i> , <i>nagC</i> , <i>ompF/asnS</i> , <i>dhaM</i> , <i>ldrC/chaA</i> , <i>narI/ECB_01206</i> , <i>topA</i> , <i>pykF</i> , <i>ynjI</i> , <i>yedW/yedX</i> , [<i>manB</i>]–[<i>cpsG</i>], <i>yegI</i> , <i>ECB_02816</i> , <i>yghJ</i> , <i>ebgR</i> , <i>infB</i> , <i>arcB</i> , <i>gltB</i> , <i>yhdG/fis</i> , <i>rpsM</i> , <i>malT</i> , <i>glpE</i> , <i>spoT</i> , <i>glmU/atpC</i> , <i>rbsD</i> –[<i>yieO</i>], <i>hslU</i> , <i>pflC</i> , <i>iclR</i> , <i>fimA</i> , <i>nadR</i>	(49)

¹Note: annotations follow conventions from <https://barricklab.org/shiny/LTEE-Ecoli/>

Table S2. Fits of several common distributions to beneficial tails of DFEs.

One-sample Kolmogorov-Smirnov test, p-values ¹						
Background	Exponential	Gamma	Weibull	Normal	LogNorm	Logistic
Anc	<10 ⁻⁴	0.035	0.289*	<10 ⁻⁴	<10 ⁻⁴	<10 ⁻⁴
2K	0.571*	0.936*	0.878*	<10 ⁻⁴	0.001	<10 ⁻⁴
15K	0.852*	0.898*	0.888*	<10 ⁻⁴	0.011	<10 ⁻⁴
Goodness-of-fit, Akaike's Information Criterion						
Background	Exponential	Gamma	Weibull	Normal	LogNorm	Logistic
Anc	-7033	-7062	-7074	-5803	-6960	-6135
2K	-4873	-4872	-4872	-4376	-4751	-4454
15K	-3518	-3516	-3516	-3186	-3449	-3233

¹Note: Asterisks indicate satisfactory fit of distribution to the observed tail of beneficial mutations based on KS test ($p > 0.05$). Bold font indicates the best-fitting distribution based on AIC criterion for each genetic background.

Table S3. Multiple linear model for prevalence of mutations in Ara+2 population.

Variable	Estimate	Std. Error	t value	$p(> t)$ ¹
Anc fitness	71.940	23.150	3.108	0.002*
2K fitness	73.480	25.290	2.906	0.004*
15K fitness	-21.170	20.100	-1.053	0.294
Target size	0.003	< 10 ⁻³	12.431	<10⁻¹⁵*

¹Note: Asterisks and bold font indicate significant effects.

Table S4: Strains from the LTEE

Population	Clone ID	Population	Clone ID
ANC (Ara -)	REL 606	ANC (Ara +)	REL607
Ara-1 (2K)	REL1164A	Ara+2 (2K)	REL1159A
Ara-1 (15K)	REL7177A	Ara+2 (15K)	REL7184A
Ara-1 (50K)	REL 11330	Ara+1 (50K)	REL 11392
Ara-2 (50K)	REL 11333	Ara+2 (50K)	REL 11342
Ara-3 (50K)	REL 11364	Ara+3 (50K)	REL 11345
Ara-4 (50K)	REL 11336	Ara+4 (50K)	REL 11348
Ara-5 (50K)	REL 11339	Ara+5 (50K)	REL 11367
Ara-6 (50K)	REL 11389	Ara+6 (50K)	REL 11370

Table S5: Primers for Recombineering:

recomb insert reverse	CTTTCTACGTGTTCCGCTTC
nadR_recomb_confirm_f	CTAACGTACCGAAAAAAGCGC
nadR_recomb_KanR_f	GCGTGTTTCGACGACTTATAATGAGGAATACGGAGGGAGATGTGTAG GCTGGAGCTGCTT
nadR_recomb_KanR_r	AACTTATTTTATACCTCGCCTTTGAGCCGTTTCATCGCGGCATATGA ATATCCTCCTTA
nadR_recomb_confirm_r	CCCGTGCAGGTTTTTTCAAAC
rbsD_recomb_confirm_f	CGGATAGACATTTAACGCTGC
rbsD_recomb_KanR_f	ATCGATGCCTTTAAGCTGAAGTAATGCTTCCATGACGGCCGTGTAG GCTGGAGCTGCTT
rbsD_recomb_KanR_r	AACTGTGGGTCAGCGAAACGTTTCGCTGATGGAGAAAAAACATATG AATATCCTCCTTA
rbsD_recomb_confirm_r	TTTCATGGTTAATCACCATGTAAAACG
lacA_recomb_confirm_f	GTGGGTCAAAGAGGCATGATG
lacA_recomb_KanR_f	CAGCGTATCAGGCAATTTTTATAATTTAAACTGACGATTCGTGTAGG CTGGAGCTGCTT
lacA_recomb_KanR_r	AACATATCAGGACGGAATGATCGCAATGAACATGCCAATGCATATG AATATCCTCCTTA
lacA_recomb_confirm_r	CTTAATTTCCGTGTTACGCTTAG
dctA_recomb_confirm_f	CATGAAAATGTCACGGAAGAAGTGA
dctA_recomb_KanR_f	CCCGCACTCGGGGAAGGGAGTGCGGGCATAAGTGATGAGAGTGTA GGCTGGAGCTGCTT
dctA_recomb_KanR_r	ATAACCTTACAAGATCTGTGGTTTTACTAAAGGACACCCTCATATGA ATATCCTCCTTA
dctA_recomb_confirm_r	GCTCAAACTTTTTAACCTTTTTGTTTCAAT
gdhA_recomb_confirm_f	TTGCTTTCCTGGGTCATTTTTTTCTT
gdhA_recomb_KanR_f	CATAAGCACAATCGTATTAATATATAAGGGTTTTATATCTGTGTAGG CTGGAGCTGCTT
gdhA_recomb_KanR_r	AGCGTAGCGCCATCAGGCATTTACAACCTAAATCACACCCCATATG AATATCCTCCTTA

gdhA_recomb_confirm_r	AATACTCATAAACGCCTGAAATTTTGC
fecE_recomb_confirm_f	CGGATGGACGATGAGTATTGGTAA
fecE_recomb_KanR_f	TTTCATTACAGTCGTGGTTTGGTTCTTACGGCCTGTGCAATGTGTAGG CTGGAGCTGCTT
fecE_recomb_KanR_r	TGCGCCGTGGTTTGTCTGGTTGCTTGTGAGAATGCGATAACATATGA ATATCCTCCTTA
fecE_recomb_confirm_r	CGATCTGCTGGCGAGAATTAT
ydiJ_recomb_confirm_f	TACTGGCATTGTCTGCTTCAAGG
ydiJ_recomb_KanR_f	ATAGCATTACAGTGCTTCCAGGGTGATTTTCCGTTTCCATAGTGTAGG CTGGAGCTGCTT
ydiJ_recomb_KanR_r	TATCGACCTACATCACAGACCGCAGGAAAGGGTCAATATACATATG AATATCCTCCTTA
ydiJ_recomb_confirm_r	GCCTTCCTGCAACACGAAAT
sspA_recomb_confirm_f	GCATATTCCATAGGAACCTGCAC
sspA_recomb_KanR_f	TAGGGACGACGTGGTGTAGCTGTGACAAATCCATACAGAGTGTAG GCTGGAGCTGCTT
sspA_recomb_KanR_r	CTGGTAGCAGTAAAAATTCTGACTATACCTGGAGGTTTTCCATATGA ATATCCTCCTTA
sspA_recomb_confirm_r	GGGTATTGCTCATTTTTTGTGTTGATT
proC_recomb_confirm_f	CCAAATTGTCATAAAGTCATCCTTTGTT
proC_recomb_KanR_f	AACAATGAATTTACGGCAGGAGTGAGGCAATGGAAAAGAGTGTA GGCTGGAGCTGCTT
proC_recomb_KanR_r	AACCGCACCGAAGTGGCGGCCTGACGTCCGGCGAAAGTCACATATG AATATCCTCCTTA
proC_recomb_confirm_r	GCTGAACCCACAAATGAGTCAC
ppc_recomb_confirm_f	CCTTAAGGATATCTGAAGGTATATTCAGAATTTG
ppc_recomb_KanR_f	ACCCTCGCGCAAAGCAGAGGGTTTGCAGAAGAGGAAGAGTGTA GGCTGGAGCTGCTT
ppc_recomb_KanR_r	ACAGGGCTATCAAACGATAAGATGGGGTGTCTGGGGTAATCATATG AATATCCTCCTTA
ppc_recomb_confirm_r	CACCGCTTTTACGTGGCTTTAT

Table S6: Primers for UMI-TnSeq Library Preparation

TnSeq_1R_01	CAAGCAGAAGACGGCATAACGAGAT GGCGAATG GTCTCGTGGGCTCGGAGAT
TnSeq_1R_02	CAAGCAGAAGACGGCATAACGAGAT CGATAGAG GTCTCGTGGGCTCGGAGAT
TnSeq_1R_03	CAAGCAGAAGACGGCATAACGAGAT TTGCGTCAG TCTCGTGGGCTCGGAGAT
TnSeq_1R_04	CAAGCAGAAGACGGCATAACGAGAT ATGATTAA GTCTCGTGGGCTCGGAGAT
TnSeq_1R_05	CAAGCAGAAGACGGCATAACGAGAT TCTTCATC GTCTCGTGGGCTCGGAGAT
TnSeq_1R_06	CAAGCAGAAGACGGCATAACGAGAT TCAGTACG GTCTCGTGGGCTCGGAGAT
TnSeq_1R_07	CAAGCAGAAGACGGCATAACGAGAT CATACCTC GTCTCGTGGGCTCGGAGAT
TnSeq_1R_08	CAAGCAGAAGACGGCATAACGAGAT GGCCAGAA GTCTCGTGGGCTCGGAGAT
TnSeq_1R_09	CAAGCAGAAGACGGCATAACGAGAT GGTCAAGT GTCTCGTGGGCTCGGAGAT

TnSeq_1R_10	CAAGCAGAAGACGGCATAACGAGATAA ACTCTCT GTCTCGTGGGCTCGGAGAT
TnSeq_1R_11	CAAGCAGAAGACGGCATAACGAGAT CTAGGTTA GTCTCGTGGGCTCGGAGAT
TnSeq_1R_12	CAAGCAGAAGACGGCATAACGAGAT TCCGGCGGG GTCTCGTGGGCTCGGAGAT
TnSeq_1R_13	CAAGCAGAAGACGGCATAACGAGAT ACGATCGCG GTCTCGTGGGCTCGGAGAT
TnSeq_1R_14	CAAGCAGAAGACGGCATAACGAGAT GTTATGCT GTCTCGTGGGCTCGGAGAT
TnSeq_1R_15	CAAGCAGAAGACGGCATAACGAGAT CAGTCGGT GTCTCGTGGGCTCGGAGAT
TnSeq_1R_16	CAAGCAGAAGACGGCATAACGAGATA AACCGCAA GTCTCGTGGGCTCGGAGAT
TnSeq_1F	CTCTTTCCCTACACGACGCTCTTCCGATCTA ANNNNN TTGGGG ACTTATCAGC CAACCT
TnSeq_2F_01	AATGATACGGCGACCACCGAGATCTACAC GGACGTAG ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_02	AATGATACGGCGACCACCGAGATCTACAC GATGCCAG ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_03	AATGATACGGCGACCACCGAGATCTACAC TCGGATGC ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_04	AATGATACGGCGACCACCGAGATCTACAC ATAACTTC ACACTCTTTCCCTACACG ACGCT
TnSeq_2F_05	AATGATACGGCGACCACCGAGATCTACAC TTCTGGCT ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_06	AATGATACGGCGACCACCGAGATCTACAC CCGGTAAC ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_07	AATGATACGGCGACCACCGAGATCTACAC CAATGCCT ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_08	AATGATACGGCGACCACCGAGATCTACAC AGATAAGA ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_09	AATGATACGGCGACCACCGAGATCTACAC CCTAATCT ACACTCTTTCCCTACACG ACGCT
TnSeq_2F_10	AATGATACGGCGACCACCGAGATCTACAC CATCTCCG ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_11	AATGATACGGCGACCACCGAGATCTACAC ACGCAGTC ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_12	AATGATACGGCGACCACCGAGATCTACAC TTCTCTTA ACACTCTTTCCCTACACG ACGCT
TnSeq_2F_13	AATGATACGGCGACCACCGAGATCTACAC GGTTCGTA ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_14	AATGATACGGCGACCACCGAGATCTACAC GTAACGAG ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_15	AATGATACGGCGACCACCGAGATCTACAC AAGAGAGT ACACTCTTTCCCTACAC GACGCT
TnSeq_2F_16	AATGATACGGCGACCACCGAGATCTACAC TGCGTAGC ACACTCTTTCCCTACAC GACGCT
TnSeq_2R	CAAGCAGAAGACGGCATAACGA

Table S7: Primer Combinations for UMI-TnSeq Samples:

	primer combination								
format (x,y)	(1R x, 2F y)			Fitness assay time point					

genetic background	t0	t1	t1	t2	t2	t3	t3	t4	t4
REL606	(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)	(7,1)	(8,1)	(9,1)
REL607	(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)	(7,2)	(8,2)	(9,2)
REL11330 (Ara-1)	(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)	(7,3)	(8,3)	(9,3)
REL11333 (Ara-2)	(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)	(7,4)	(8,4)	(9,4)
REL11364 (Ara-3)	(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)	(7,5)	(8,5)	(9,5)
REL11336 (Ara-4)	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)	(7,6)	(8,6)	(9,6)
REL11339 (Ara-5)	(1,7)	(2,7)	(3,7)	(4,7)	(5,7)	(6,7)	(7,7)	(8,7)	(9,7)
REL11389 (Ara-6)	(1,8)	(2,8)	(3,8)	(4,8)	(5,8)	(6,8)	(7,8)	(8,8)	(9,8)
REL11392 (Ara+1)	(1,9)	(2,9)	(3,9)	(4,9)	(5,9)	(6,9)	(7,9)	(8,9)	(9,9)
REL11342 (Ara+2)	(1,10)	(2,10)	(3,10)	(4,10)	(5,10)	(6,10)	(7,10)	(8,10)	(9,10)
REL11345 (Ara+3)	(1,11)	(2,11)	(3,11)	(4,11)	(5,11)	(6,11)	(7,11)	(8,11)	(9,11)
REL11348 (Ara+4)	(1,12)	(2,12)	(3,12)	(4,12)	(5,12)	(6,12)	(7,12)	(8,12)	(9,12)
REL11367 (Ara+5)	(1,13)	(2,13)	(3,13)	(4,13)	(5,13)	(6,13)	(7,13)	(8,13)	(9,13)
REL11370 (Ara+6)	(1,14)	(2,14)	(3,14)	(4,14)	(5,14)	(6,14)	(7,14)	(8,14)	(9,14)

Table S8. Primers used in the INSeq approach.

Name	sequence
MS-custom-adaptor-F	5'-TTCCCTACACGACGCTCTTCCGATCTNN-3'
MS-custom-adaptor-R	5'-AGATCGGAAGAGCGTCGTGTAGGGAA-3'
HS-custom-adaptor-F	5'-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNN-3'
HS-custom-adaptor-R	5'-CTGTCTCTTATACACATCTGACGCTGCCGACGA-3'
MS-junction-PCR-F	5'-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGNNNNNNNTTCCCTACACGACGCTCTTCGATCT-3
MS-junction-PCR-R	5'-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNNNNNNNAGACCGGGGACTTATCATCCAACCTGT-3

HS-junction-PCR-F	5'-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG-3'
HS-junction-PCR-R	5'-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGNNNNNAGACCGGGGACTTATCAT CCAACCTGT-3'

Data Table Titles S1 to S5:

Data S1: Counts from fitness assays based on direct competitions

Data S2: List of all genes differentially essential over evolution

Data S3: Homologous gene groups in the ancestral strain ANC

Data S4: Fitness effects of *wecA* insertion mutants

Data S5: Essentiality and expression levels of *kdsB* genes

References and Notes

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